

# **Bio-Cyber Hybrid Neuromorphic Chemical Sensor for Real-Time Household Gas Detection**

*A Project Report*

*Submitted to APJ Abdul Kalam Technological University*

*In partial fulfillment of requirements for award of degree*

*Bachelor of Technology*

in

**COMPUTER SCIENCE AND ENGINEERING**

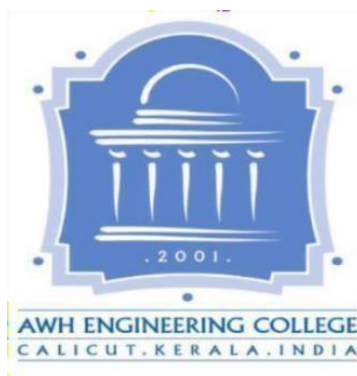
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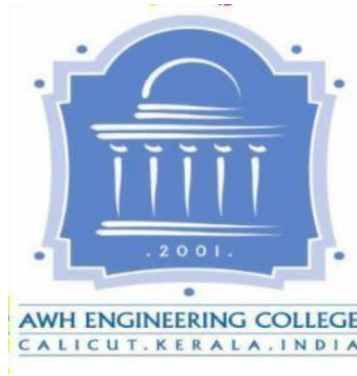
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**CERTIFICATE**

This is to certify that the report entitled “**Bio-Cyber Hybrid Neuromorphic Chemical Sensor for Real-Time Household Gas Detection**” submitted by **FATHIMA SHIFA (AWH22CS018), NIBIN NIHAD (AWH22CS043), SALMAN NV (AWH22CS049), MUHAMMED INSAMAM CP (AWH22CS038)** to the APJ Abdul Kalam Technological University in partial fulfillment of the B.Tech degree in Computer Science and Engineering is a bonafide record of the project work carried out by him under our guidance and supervision. This report in any form has not been submitted to any other University or Institute for any purpose.

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# DECLARATION

We hereby declare that the project report “**Bio-Cyber Hybrid Neuromorphic Chemical Sensor for Real-Time Household Gas Detection**”, submitted for partial fulfillment of the requirements for the award of degree of Bachelor of Technology of the APJ Abdul Kalam Technological University, Kerala is a bonafide work done by us under supervision of **Ms. Gopika P.** This submission represents our ideas in our own words and where ideas or words of others have been included, we have adequately and accurately cited and referenced the original sources. We also declare that we have adhered to ethics of academic honesty and integrity and have not misrepresented or fabricated any data or idea or fact or source in our submission. We understand that any violation of the above will be a cause for disciplinary action by the institute and/or the University and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been obtained. This report has not been previously formed the basis for the award of any degree, diploma or similar title of any other University.

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# ABSTRACT

Gas leaks, especially involving LPG and biogas, pose severe risks in households. Conventional gas detection systems often rely on binary threshold detection and lack real-time intelligence or adaptability. This project introduces a beginner-accessible, bio-cyber hybrid system that mimics the human olfactory system using Spiking Neural Networks (SNNs). By integrating MQ-series sensors (MQ-2 and MQ-5) with an ESP32 microcontroller, the system captures gas concentrations and encodes them into spike-time representations. These spike patterns are processed using a lightweight SNN simulated in Brian2, enabling intelligent classification and early warning. The system also includes a mobile application (built using Flutter) that receives real-time gas data and triggers a beeping alarm if leakage is detected. Designed for adorability and ease of implementation, the solution provides energy-efficient operation for household safety. This approach bridges advanced neuromorphic computing with practical embedded systems, overing a scalable path toward intelligent gas detection in domestic environments.

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# CHAPTER 1

## INTRODUCTION

### 1.1 OVERVIEW

Gas leakage detection is a critical aspect of household and industrial safety. Conventional gas detection systems rely on threshold-based methods that trigger alarms when gas concentrations exceed a certain limit. Although functional, these systems suffer from limitations such as high power consumption, limited adaptability to mixed gas environments, and poor scalability. With the increasing emphasis on smart homes and IoT-based safety systems, there is a strong demand for intelligent, low-power, and biologically inspired sensing systems that can detect gas leaks more efficiently and reliably.

Neuromorphic engineering—a discipline inspired by the biological structure and functioning of the human brain—has emerged as a promising paradigm for designing energy-efficient and adaptive sensors. In particular, the human olfactory system, capable of identifying thousands of odor molecules using neural spikes, provides an excellent biological model for gas detection applications. By combining neuromorphic computing techniques with chemical sensing, it is possible to create a hybrid system capable of mimicking the brain’s event-driven processing approach.

This project, titled “**Bio-Cyber Hybrid: Neuromorphic Chemical Sensor for Real-Time Household Gas Detection,**” aims to design and develop a cost-effective, brain-inspired gas sensing system capable of detecting LPG and biogas leaks in domestic environments.

### 1.2 BACKGROUND AND MOTIVATION

Household gas leaks, primarily caused by LPG and biogas leakage, remain one of the major sources of fire hazards and accidental explosions in residential areas. Traditional systems such as MQ-based gas detectors or semiconductor sensors provide basic alerts but lack intelligence, adaptability, and energy efficiency.

Recent advancements in **neuromorphic computing** and **spiking neural networks (SNNs)** enable hardware systems to emulate brain-like learning and sensory processing at a fraction of the energy cost of conventional digital systems. By leveraging this approach, the project aims to build a **bio-cyber hybrid system** that replicates the olfactory neural pathways, enhancing accuracy and

response time while minimizing power consumption.

The motivation behind this research lies in the need for an **affordable, efficient, and real-time** gas detection solution that can be deployed in households, small industries, and smart kitchens to prevent catastrophic incidents and safeguard human lives.

### 1.3 PROBLEM DEFINITION

Traditional gas detectors function by comparing sensor outputs to predefined thresholds. While effective for specific gases, this method is limited in:

- Distinguishing between multiple gas types.
- Adapting to varying environmental conditions.
- Reducing false alarms due to humidity or temperature fluctuations.

To overcome these limitations, this project proposes a hybrid model that integrates **chemical gas sensors (MQ-series)** with a **neuromorphic SNN model**. The SNN processes gas concentration data in the form of **spike-time encodings**, replicating the information processing of biological neurons. The system generates both **local alerts (buzzer)** and **mobile notifications (via a Flutter app)**, ensuring users receive warnings even when away from the source.

### 1.4 OBJECTIVES

The major objectives of the proposed work are as follows:

1. To design and develop a **bio-inspired gas detection system** that mimics human olfactory processing.
2. To implement a **Spiking Neural Network (SNN)** model for real-time classification of gas concentration levels.
3. To integrate **MQ-series sensors** with **ESP32 microcontroller** for efficient signal acquisition.
4. To develop a **mobile alert application** using **Flutter** for remote monitoring and instant notifications.
5. To evaluate the system based on **accuracy, latency, and power consumption**, ensuring it is suitable for household deployment.

## 1.5 SCOPE

The scope of this project includes the complete design and implementation of a low-cost neuromorphic gas detector for domestic environments. The system:

- Focuses on detecting **LPG and biogas** leaks.
- Operates using **low-power hardware** suitable for continuous operation.
- Employs **event-based spike encoding** for gas concentration data.
- Features **real-time communication** between sensor and mobile app via **MQTT or Firebase**.
- Provides both **visual and audio alerts** to warn users.

This project does not cover complex multi-gas classification or large-scale industrial deployment but sets the foundation for such extensions.

## 1.5 APPLICATIONS

The proposed system can be deployed in various environments:

- **Smart homes:** For early detection of LPG or biogas leaks.
- **Restaurants and food industries:** Continuous safety monitoring in kitchens.
- **Biogas plants:** Detection of methane leaks to prevent explosions.
- **Smart cities and IoT frameworks:** As a part of a distributed safety monitoring system.
- **Educational research:** A low-cost platform to explore neuromorphic sensing principles.

## **CHAPTER 2**

### **LITERATURE SURVEY**

#### **2.1 GAS LEAK REAL-TIME DETECTION AND VOLUME FLOW QUANTIFICATION BASED ON INFRARED IMAGING AND ADVANCED ALGORITHMS**

##### **2.1.1 AIM**

This paper proposes a groundbreaking solution to alleviate funding shortages for infrastructure. The primary aim of this research is to develop an intelligent, real-time, and cost-effective system for the detection and quantification of gas leaks using only an infrared imaging device combined with advanced algorithms. Detecting gases is a challenging task because gases are semi-transparent and have irregular diffusion patterns, especially when their flow rates are very small. The study focuses on designing a software-based solution capable of identifying and quantifying trace gas leaks in the mid-infrared region without relying on bulky or expensive instruments such as Fourier-transform infrared spectrometers or laser-based systems. The objective is to automate gas identification, segmentation, and flow quantification, making the process more affordable and suitable for real-world industrial applications such as safety monitoring and environmental protection.

##### **2.1.2 METHODOLOGY**

The proposed method integrates an infrared camera with optimized deep learning and image-processing algorithms to automatically detect and measure gas leaks in real time. Initially, the system captures infrared images of gas plumes using a thermal camera that operates in the 3.2–3.5  $\mu\text{m}$  wavelength range. These images are then processed using a modified version of the DeepLabV3+ semantic segmentation network. To enhance speed and efficiency, the original Xception backbone of DeepLabV3+ is replaced with MobileNetV2, which makes the model lightweight and capable of real-time operation. Furthermore, two specialized attention mechanisms—Coordinate Attention (CA) and Efficient Channel Attention (ECA)—are incorporated to improve the network's sensitivity in recognizing blurred and low-contrast gas edges. After segmentation, an optimized K-means clustering algorithm is applied to automatically

determine the region of interest (ROI) by removing irrelevant background and refining the boundaries of the gas plume. Once the ROI is established, the system quantifies the gas flow using a combination of the radiative transfer model and the optical flow method. The radiative transfer model estimates the gas column density from the infrared radiation data, while the optical flow method calculates the velocity of gas movement between successive frames. Finally, by integrating these results with physical parameters such as pressure, temperature, and molecular weight, the actual gas volume flow rate is calculated. Experiments using butane gas as the target were conducted to validate the approach, and the results showed that the proposed system achieved a detection limit as low as 0.01 L/min, with rapid processing of approximately 0.03 seconds per frame.

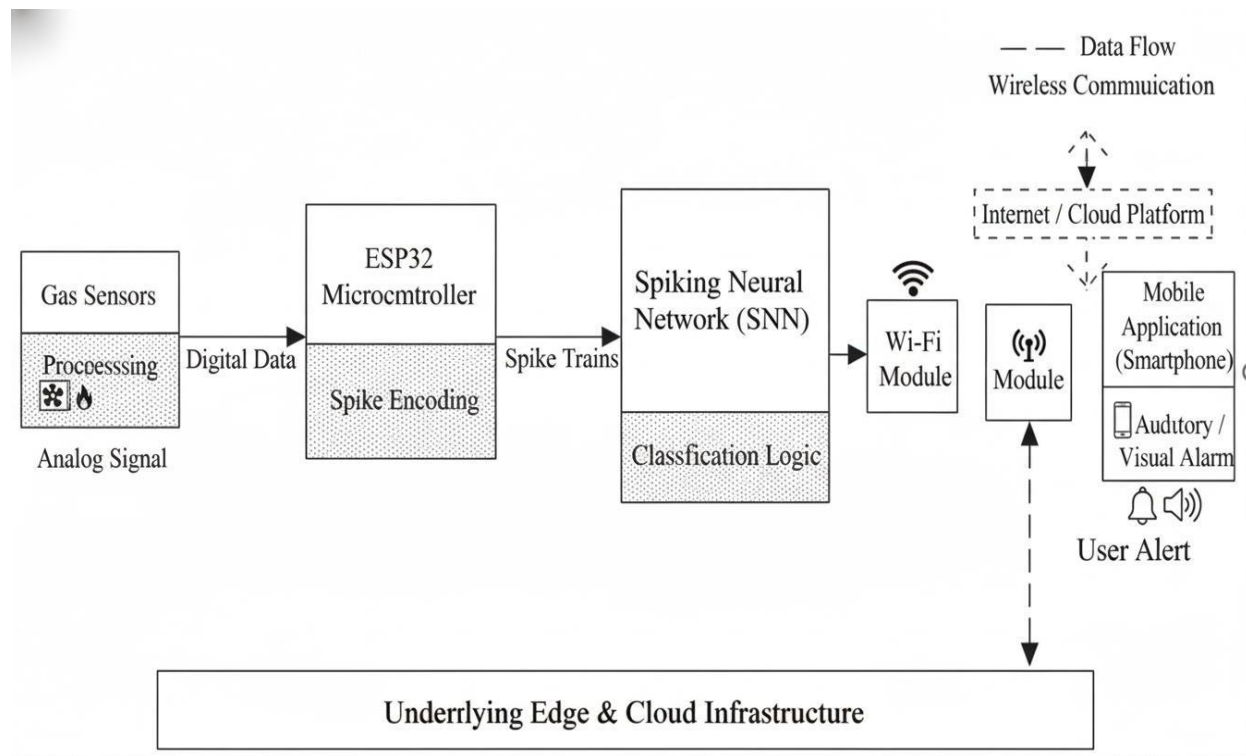


Fig 2.1 Workflow of 1<sup>st</sup> survey paper

### 2.1.3 ADVANTAGES

- **Low-Cost Implementation:**

The system requires only a single infrared camera and does not depend on expensive equipment such as FTIR spectrometers or laser-based systems.

- **Real-Time Operation:**

The modified DeepLabV3+ network with the MobileNetV2 backbone enables fast process

achieving results in approximately 0.03 seconds per frame.

- **High Sensitivity:**

Capable of detecting extremely small gas flow rates down to 0.01 L/min, which is comparable to the performance of advanced laboratory systems.

## 2.1.4 DISADVANTAGES

- **Environmental Dependence:**

Performance is affected by temperature difference ( $\Delta T$ ) between the gas and the background, and the distance from the camera to the gas source.

- **Limited Detection Range:**

For weak leaks, the gas and background signals become indistinguishable beyond a few meters (typically 2–5 m).

- **Angle Sensitivity:**

The accuracy drops significantly if the infrared camera's line of sight deviates more than 30° from the gas flow direction

## **2.2 NDAMA: A NOVEL DEEP AUTOENCODER AND MULTIVARIATE ANALYSIS APPROACH FOR IOT-BASED METHANE GAS LEAKAGE DETECT**

### **2.2.1 AIM**

The primary aim of this research is to design an intelligent, reliable, and real-time methane gas leakage detection system that integrates deep learning techniques with IoT (Internet of Things) infrastructure. Methane is a highly flammable and potent greenhouse gas, and its leakage poses both safety and environmental risks. Traditional detection systems based on threshold sensors often suffer from false alarms, delayed responses, and inability to handle complex, multivariate sensor data. Therefore, the paper introduces NDAMA (Novel Deep Autoencoder and Multivariate Analysis) — a hybrid framework that combines deep autoencoders for feature extraction and multivariate statistical analysis for robust leak identification. The objective is to achieve early detection, high accuracy, and real-time alerting of methane leaks in industrial IoT environments, ensuring better safety and sustainability.

### **2.2.2 METHODOLOGY**

The methodology proposed in this research combines deep learning, multivariate statistical analysis, and IoT-based sensing to create an intelligent and reliable framework for detecting methane gas leakages in real time. The approach, called NDAMA (Novel Deep Autoencoder and Multivariate Analysis), consists of several systematic stages — from data collection to model deployment — ensuring both accuracy and practicality in real-world environments.

The first stage involves IoT-based data collection, where multiple sensors are deployed in the target environment to continuously monitor parameters related to gas leakage. These sensors measure methane concentration ( $\text{CH}_4$ ) along with other environmental features such as temperature, humidity, wind speed, and pressure. The use of IoT technology allows real-time data transmission from various points in an industrial setup to a central server or cloud platform. This stage ensures continuous monitoring and creates a large dataset for model training and analysis.

The second stage focuses on data preprocessing, which is a crucial step to ensure data quality and reliability. Since raw sensor data may contain noise, missing values, or measurement errors due to environmental variations, preprocessing techniques are applied to clean and normalize the data. In the NDAMA model, a Deep Autoencoder (DAE) is used to identify and remove outliers. The



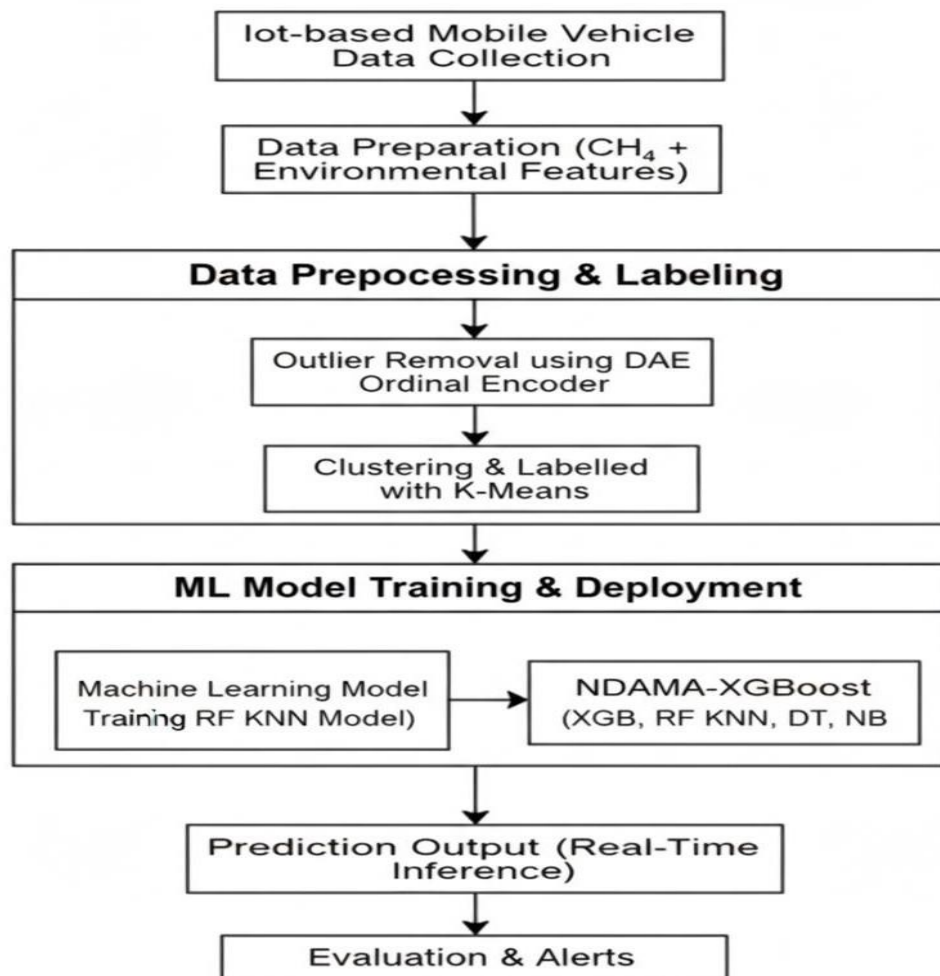
autoencoder learns the normal operating pattern of sensor readings and automatically detects anomalies that deviate from these patterns. Additionally, an Ordinal Encoder (OE) or normalization method is used to scale features into a comparable range, making them suitable for machine learning algorithms. This step ensures that irrelevant noise or irregular values do not affect the accuracy of the subsequent analysis.

The third stage, data labeling, prepares the dataset for supervised learning. Methane concentration ( $\text{CH}_4$ ) is used as the primary feature for categorizing the data into different leakage levels such as low, medium, or high. To achieve this, the K-means clustering algorithm is applied, which classifies the sensor data into clusters based on similarities in methane concentration levels. This approach automatically assigns labels to data points without the need for manual labeling, reducing the dependence on human intervention.

The fourth stage, predictive analysis, involves training machine learning models to detect and classify gas leakages based on the preprocessed and labeled data. Several classifiers such as XGBoost, Random Forest (RF), K-Nearest Neighbors (KNN), Decision Tree (DT), and Naïve Bayes (NB) are implemented and compared to identify the best-performing model. The models are trained using cross-validation and grid search techniques for optimal hyperparameter tuning. The combination of the Deep Autoencoder (for feature extraction), Ordinal Encoder (for data normalization), and XGBoost (for classification) — referred to as DA-OE-XGBoost — was found to yield the highest accuracy, achieving up to 99.5% detection accuracy. This integration allows the system to effectively recognize both linear and nonlinear relationships between multiple environmental variables, thereby improving detection reliability.

The final stage is deployment and real-time alerting. The trained NDAMA model is integrated into the IoT platform for live data monitoring. Incoming data from sensors are continuously fed into the trained model, which analyzes it in real time to determine whether a gas leak has occurred. When the system identifies an abnormal pattern that indicates a methane leak, it triggers automatic alerts through cloud dashboards, email notifications, or mobile apps, allowing immediate action to prevent accidents. The IoT integration also enables remote monitoring, periodic model retraining, and system scalability across multiple industrial locations.

Overall, the NDAMA methodology represents a complete pipeline from data collection to intelligent decision-making. It successfully merges deep learning's capability for nonlinear feature extraction with multivariate analysis's strength in handling correlated environmental parameters. By leveraging IoT infrastructure, it ensures continuous, accurate, and early detection of methane leaks, making it a powerful solution for industrial safety and environmental monitoring.

Fig 2.2 Workflow of 2<sup>nd</sup> survey paper

### 2.2.3 ADVANTAGES

- **High Detection Accuracy:**

The proposed NDAMA framework (Deep Autoencoder with Multivariate Analysis) achieves high accuracy and reliability in detecting methane gas leaks by combining deep learning with statistical analysis.

- **Early and Real-Time Detection:**

The IoT-enabled architecture allows continuous monitoring and early detection of gas leaks before they reach hazardous levels.

- **Robust Feature Extraction:**

The deep autoencoder effectively extracts hidden nonlinear patterns and correlations among multiple environmental parameters such as temperature, pressure, and gas concentration.

## 2.2.4 DISADVANTAGES

- **High Initial Computational Cost:**

The deep autoencoder model requires substantial training data and computational power during the training phase, which may not be feasible for small facilities.

- **Dependence on Sensor Quality:**

The accuracy of detection heavily depends on the precision and calibration of IoT gas sensors; low-quality sensors can degrade performance.

- **Limited Generalization Without Recalibration:**

The model may need retraining or recalibration when applied to new environments, sensor types, or gas compositions.

## **2.3 A FRAMEWORK TO SURVEY A REGION FOR GAS LEAKS USING AN UNMANNED AERIAL VEHICLE**

### **2.3.1 AIM**

The main aim of this research is to develop a systematic and reliable framework for detecting and mapping gas leaks using an Unmanned Aerial Vehicle (UAV) equipped with gas sensing and environmental monitoring instruments. The framework is designed to automate the process of surveying large geographical areas for methane or other gas emissions while minimizing human effort and risk. Traditional ground-based detection methods are often slow, labor-intensive, and unsafe, especially in complex terrains or hazardous industrial zones. To overcome these challenges, the proposed approach enables UAVs to collect real-time methane concentration data along with wind and positional information, and then convert these measurements into a survey map that identifies regions likely to contain gas leaks and regions that can be safely declared leak-free. The research specifically aims to provide a statistically grounded and auditable method for interpreting UAV data through the concept of Upwind Survey Regions (USRs) and a data fusion algorithm, ensuring that gas detection is both accurate and scientifically verifiable. Ultimately, the goal is to enhance the efficiency, safety, and reliability of large-scale gas leak inspection and monitoring operations.

### **2.3.2 METHODOLOGY**

The proposed methodology in this paper presents a structured framework that enables surveying regions for methane gas leaks using an Unmanned Aerial Vehicle (UAV) equipped with advanced sensing and mapping capabilities. The key goal is to convert UAV methane concentration measurements into a survey map that indicates areas likely to contain gas leaks and those that can be confidently cleared. The framework consists of two main components — Upwind Survey Region (USR) Mapping and USR Fusion, which together enable systematic leak detection and classification.

The first step in the methodology involves data collection using a UAV equipped with methane sensors. The UAV carries an Open-Path Laser Spectrometer (OPLS) sensor, GPS, and anemometer to measure methane concentration, location, wind direction, and wind speed. As the UAV flies in a sweeping flight pattern downwind of the area being surveyed, it collects in-situ methane

threshold  $C_{\min}$ .

Next, the framework introduces the concept of an Upwind Survey Region (USR) to relate each UAV measurement to a corresponding region on the ground. A USR represents the area upwind of the UAV's position that could have contributed to the measured concentration. If a high methane concentration is recorded, the USR likely contains a leak; if a low concentration is recorded, the USR is likely clear of leaks. The shape and dimensions of each USR are modeled as ellipsoidal regions based on environmental factors such as wind speed, turbulence, and UAV altitude. These parameters are pre-calculated using the Quick Urban and Industrial Complex (QUIC) simulation tool, which estimates gas dispersion behavior and stores USR models in a database. This step ensures that appropriate USR models are selected under varying atmospheric and flight conditions. Once each measurement is mapped to its USR on the ground, the USR Mapping process records how many detections and non-detections cover each subregion of the survey area. This mapping results in multiple overlapping USRs, some indicating detections and others non-detections. Because these measurements may conflict due to gas plume turbulence and wind variability, a statistical fusion process is required to interpret the results.

The USR Fusion stage statistically combines information from overlapping USRs to classify each subregion into one of three categories:

- R(gas): areas that are likely to contain gas leaks,
- R(clear): areas that can be confidently declared free of gas leaks, and
- R(unknown): areas where data is insufficient for a reliable decision.

The fusion process uses a binomial statistical model to calculate the probability (p-value) of each subregion being leak-free or leak-prone, considering multiple cross-flights and measurement repetitions. It also employs false positive and false negative control to minimize misclassification. The statistical thresholds ensure that only areas with strong evidence are marked as leak-free, making the survey conservative and reliable.

Finally, all classified subregions are combined into a survey map, which visually partitions the entire region into R(gas), R(clear), and R(unknown) areas. This map provides gas companies with an actionable tool to prioritize inspection and maintenance, focusing human patrols only on areas identified as probable leak zones.

In summary, the methodology integrates UAV-based sensing, aerodynamic modeling, statistical data fusion, and simulation-assisted calibration to achieve accurate, auditable, and efficient gas leak surveys. This structured approach ensures a balance between safety, reliability, and coverage efficiency in real-world gas monitoring operations.

## UAV-BASED METHANE MEASUREMENT AND LEAK DETECTION

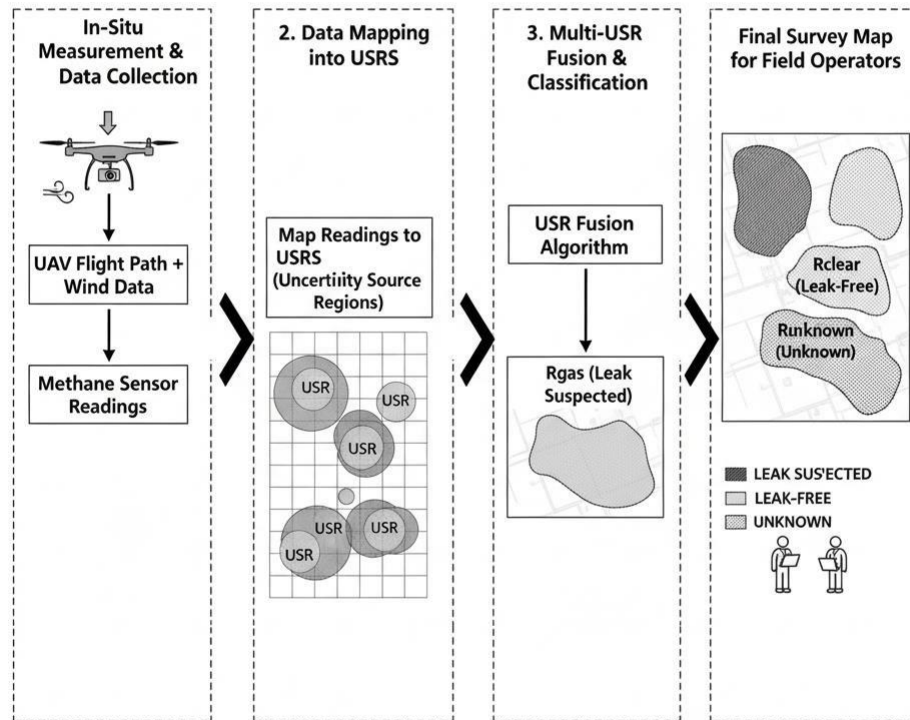


Fig 2.3 Workflow of 3<sup>rd</sup> survey paper

### 2.3.3 ADVANTAGES

- **Wide-Area Coverage:**

The UAV-based system can efficiently survey large or difficult-to-access regions, such as industrial sites, pipelines, and remote gas fields, which would be unsafe or time-consuming for human inspection.

- **Reduced Human Risk:**

The framework eliminates the need for personnel to enter hazardous or high-pressure gas zones by using autonomous aerial data collection.

- **Accurate Leak Localization:**

The use of the Upwind Survey Region (USR) model allows the UAV to estimate potential gas leak origins based on wind direction and dispersion characteristics, leading to accurate source localization.

### 2.3.4 DISADVANTAGES

- **Dependence on Environmental Conditions:**

The accuracy of leak detection depends heavily on wind stability, temperature, and turbulence. Sudden changes in weather can distort gas plume behavior and lead to uncertainty in USR estimation.

- **High Computational Demand:**

Generating and processing multiple overlapping USRs, as well as performing statistical fusion, can be computationally intensive, requiring strong onboard or ground-based processing capabilities.

- **Sensor Limitations:**

The methane detection sensitivity is constrained by the accuracy and response time of the onboard Open-Path Laser Spectrometer (OPLS). Low-concentration or intermittent leaks may go undetected

## 2.4 LOW LIMIT OF DETECTION GAS DENSITY SENSING WITH A DIGITALLY PI-CONTROLLED MICROCANTILEVER

### 2.4.1 AIM

The aim of this research is to develop a **new, highly sensitive and low-noise platform** for **detecting gas density variations** using a **digitally controlled uncoated microcantilever**.

The system seeks to:

- Achieve **real-time detection** of very small changes in gas properties such as **density and viscosity**.
- Reach **unprecedented responsivity** (sensitivity to density changes) and **ultra-low limits of detection (LoD)** — much lower than previous resonant sensors.
- Avoid the need for **chemical coatings or surface functionalization**, which often complicate fabrication and reduce stability.
- Provide a **general, stable, and tunable** sensing method that can be extended to **other physical sensing tasks**, including mass, viscosity, and viscoelasticity detection.

In essence, the paper's goal is to create a **digital, phase-locked-loop (PLL)-controlled microcantilever system** capable of detecting **minute gas density variations (down to  $3.5 \times 10^{-4} \text{ kg/m}^3$ )** with high precision and low noise.

### 2.4.2 METHODOLOGY

The proposed methodology integrates experimental design, analytical modeling, and digital control to achieve high-precision gas density sensing using an uncoated microcantilever. The developed system consists of two main modules — an **optomechanical unit** and an **electronic control unit**. In the optomechanical unit, a **silicon microcantilever** is placed inside a sealed chamber filled with gas, where it is **excited and its deflection is detected optically** using a laser beam reflected onto a four-quadrant photodiode. The electronic module includes **direct digital synthesizers (DDS)**, **analog-to-digital converters (A/D)**, a **dsPIC microcontroller**, and a **Raspberry Pi interface**, which together form a **digitally controlled phase-locked loop (PLL)** system. The cantilever is driven to oscillate at a controlled frequency, and the **phase difference ( $\phi$ )** between the excitation and deflection signals is **digitally imposed** by the system. The **quadrature (Q) component** of the signal acts as an **error parameter** in a **proportional–integral (PI) controller**, which continuously adjusts the oscillation frequency to maintain a stable phase



relationship.

Any variation in the surrounding gas density or viscosity causes a small shift in the **resonance frequency** of the cantilever, which the system detects in real time. A **theoretical analytical model** was developed to describe the cantilever's dynamic behavior as a **damped harmonic oscillator**, incorporating the effects of added mass and damping due to the gas environment. This model helps predict how changes in gas properties influence frequency and amplitude. The equations derived from this model are used to calculate the system's **responsivity** to density variations and optimize the operating phase for maximum sensitivity. Experimental validation was carried out using different gases such as air, nitrogen, and helium, under varying pressures. The results were compared with model predictions to confirm the accuracy of the system. Additionally, **Allan variation analysis** was used to evaluate frequency stability and noise performance, ensuring that the minimum detectable frequency shift could be resolved. Finally, the **limit of detection (LoD)** was experimentally determined by monitoring the smallest measurable frequency changes corresponding to minute density variations, confirming the system's superior sensitivity and performance.

## Microcantilever Gas Density Sensor System

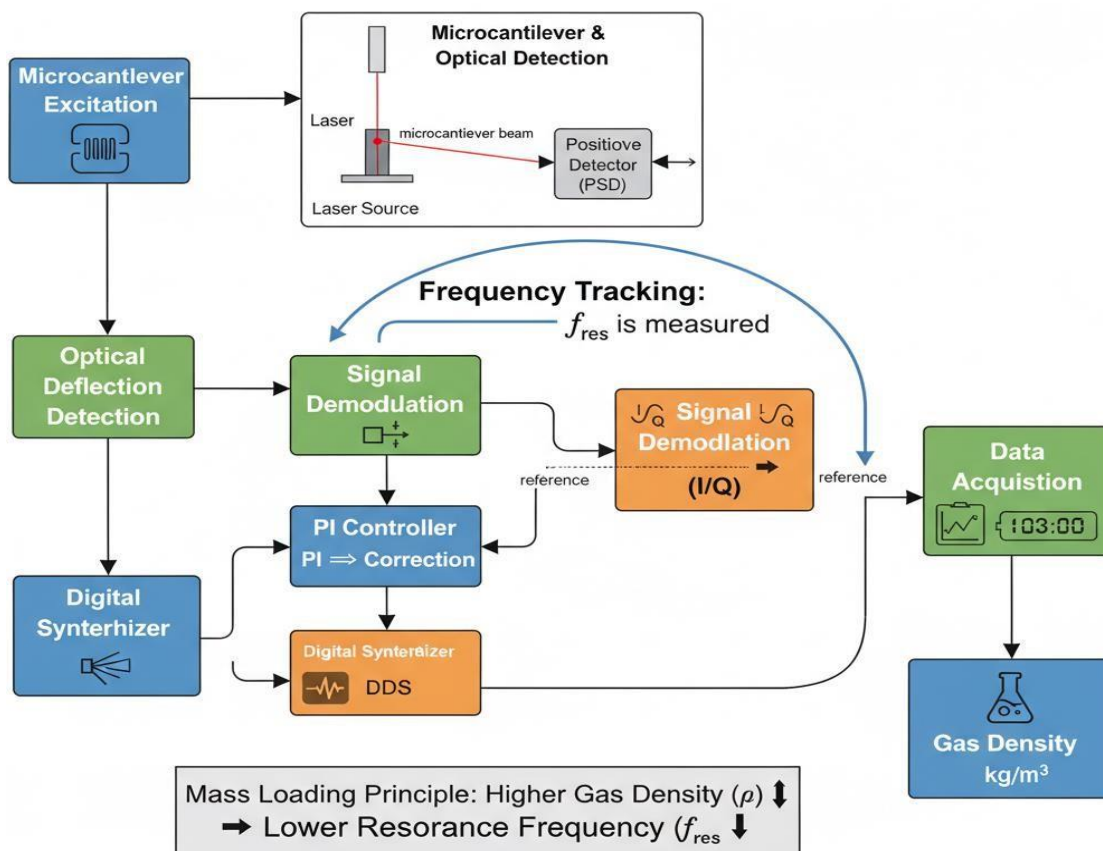


Fig 2.4 Workflow of 4<sup>th</sup> survey paper

### 2.4.3 ADVANTAGES

- **Extremely High Sensitivity:**

The system achieves a record-low limit of detection of  $3.5 \times 10^{-4} \text{ kg/m}^3$  in air, which is one to two orders of magnitude better than any previous uncoated resonant gas density sensor. This allows the detection of even the tiniest gas density variations.

- **No Need for Chemical Coating:**

Unlike traditional gas sensors that require chemical functionalization, this sensor uses an uncoated microcantilever. This makes fabrication simpler, avoids contamination or degradation over time, and enhances long-term stability and reliability.

- **Digital Precision and Control:**

The use of direct digital synthesis (DDS) and phase-locked loop (PLL) control ensures highly precise frequency generation and measurement with very low noise levels. This digital design improves accuracy, reduces drift, and enhances measurement repeatability.

### 2.4.4 DISADVANTAGES

- **Environmental Sensitivity:**

The system's performance can be influenced by temperature fluctuations, humidity, and laser intensity changes, which may cause frequency drift or noise. Proper environmental control or compensation methods are required for accurate results.

- **Precise Optical Alignment Required:**

The laser-based optical detection setup demands highly accurate alignment of the laser and cantilever, making the system delicate and difficult to operate in non-laboratory or field environments.

- **Complex Controller Tuning:**

The PI controller must be carefully tuned for each operating condition. Incorrect gain settings can cause instabilities, delayed responses, or excessive noise, reducing sensor performance.

## 2.5 MIXED GAS DETECTION AND TEMPERATURE COMPENSATION BASED ON PHOTOACOUSTIC SPECTROSCOPY

### 2.5.1 AIM

The main aim of this study is to develop a **highly accurate and reliable mixed gas detection system** using **Photoacoustic Spectroscopy (PAS)**, enhanced with **machine learning-based classification and temperature compensation algorithms**. Specifically, the research seeks to:

- **Improve the accuracy of gas component identification and concentration measurement** in complex mixed gas environments.
- **Overcome the limitations of conventional SVM classification** by proposing a **hybrid KNN-SVM algorithm** that better handles ambiguous data and enhances classification accuracy.
- **Address the temperature sensitivity** of PAS detection systems through a **WOA-BP (Whale Optimization Algorithm–Backpropagation)** neural network model for **temperature compensation**.

The goal is to create a **comprehensive, intelligent detection framework** capable of real-time and precise detection of trace gases under variable temperature conditions, ensuring **stability, accuracy, and fast convergence**.

### 2.5.2 METHODOLOGY

The methodology combines **experimental PAS gas detection** with **machine learning algorithms and neural network-based compensation models**. A **photoacoustic spectroscopy setup** is constructed using an **infrared laser light source**, a **PA201 photoacoustic detector cell**, and a **cantilever-type pressure sensor** to detect acoustic signals produced when the target gases absorb modulated light. The gas delivery system mixes different gases—**C<sub>2</sub>H<sub>2</sub>, NO<sub>2</sub>, and SF<sub>6</sub>**—with nitrogen in controlled ratios to generate various test samples. The **photoacoustic signals** are collected, amplified, and processed using a **DSP unit** and **LabView interface** to obtain amplitude, frequency, and absorption spectra data.

The **KNN-SVM hybrid classification algorithm** is then applied for qualitative gas identification. The system first uses the **SVM** model to determine hyperplanes that separate gas classes, while **KNN** reclassifies ambiguous data points near the decision boundary using a distance-based voting mechanism. This two-layer classification process improves the recognition accuracy of mixed gases and minimizes misclassification of overlapping data.

For **temperature compensation**, the study introduces a **WOA-BP neural network model**. The **Whale Optimization Algorithm (WOA)** optimizes the weights and biases of a **Backpropagation (BP)** neural network by simulating whale hunting behavior to achieve global optimization. The trained model predicts and compensates for detection errors caused by temperature variations. The performance of WOA-BP is compared with **standard BP**, **GA-BP (Genetic Algorithm optimized BP)**, and **PSO-BP (Particle Swarm Optimization BP)** using metrics such as **R<sup>2</sup>**, **MAE**, and **RMSE**. The results show that WOA-BP offers faster convergence, higher precision, and lower prediction errors, effectively stabilizing PAS detection under temperature fluctuations.

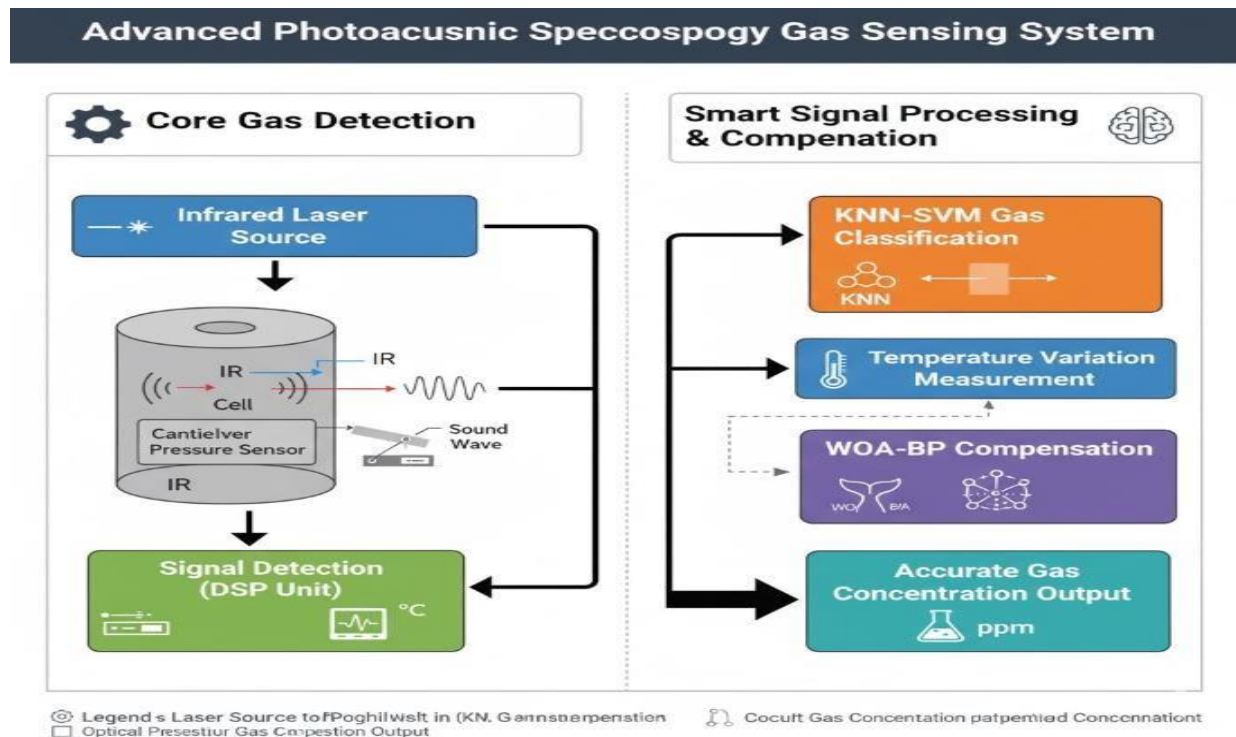


Fig 2.5 Workflow of 5<sup>th</sup> survey paper

### 2.5.3 ADVANTAGES

- **High Classification Accuracy:**

The hybrid KNN-SVM algorithm achieves 99.167% accuracy and an AUC of 99.375%, significantly improving gas identification compared to traditional SVM methods.

- **Enhanced Data Handling:**

Ambiguous or overlapping data points are reclassified using KNN voting, making the algorithm more robust and reliable in real-world mixed gas scenarios.

- **Improved Temperature Compensation:**

The WOA-BP model provides excellent compensation for temperature-related

errors, achieving  $R^2 = 97.89\%$ ,  $MAE = 1.4868$ , and  $RMSE = 2.0416$ , ensuring highly stable gas concentration measurements.

## 2.5.4 DISADVANTAGES

- **High Computational Demand:**

The combined KNN-SVM and WOA-BP models require significant processing power and training time, which may not be ideal for low-resource embedded systems.

- **Need for Large Training Datasets:**

The accuracy of machine learning models depends on ample and diverse data, making it challenging to generalize when sample size or variety is limited.

- **Temperature Dependency of PAS System:**

Even with compensation, extreme temperature changes can still affect acoustic resonance and light absorption, requiring frequent recalibration.

## 2.6 NEUROMORPHIC WIRELESS SPLIT COMPUTING WITH MULTI-LEVEL SPIKES

### 2.6.1 AIM

The main aim of this research is to **develop and analyze a neuromorphic wireless split computing system** that uses **multi-level spiking neural networks (SNNs)** to enhance the efficiency, accuracy, and scalability of artificial intelligence (AI) processing at the edge of wireless networks.

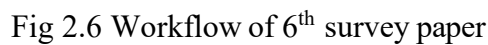
Traditional AI systems rely on GPUs and large centralized servers, which are energy-intensive and unsuitable for real-time edge applications. This study aims to overcome those limitations by combining **neuromorphic computing**—which mimics the human brain’s energy-efficient way of processing information—with **wireless split computing**, where computation is shared between two devices (a transmitter and a receiver).

### 2.6.2 METHODOLOGY

The study adopts an experimental and simulation-based methodology to design, implement, and evaluate a neuromorphic wireless split computing system using multi-level spiking neural networks (SNNs). The proposed system divides an SNN into two functional parts—an encoding SNN located at the transmitter and a decoding SNN at the receiver. The transmitter collects event-driven sensory data from a neuromorphic camera and processes it into spike signals, while the receiver reconstructs and interprets these spikes to perform the final inference task. Unlike conventional binary SNNs, the system employs a multi-level leaky integrate-and-fire (M-LIF) neuron model, where each spike carries multiple bits of information instead of a single bit. This design allows for richer data transmission without significantly increasing energy consumption.

To enable efficient wireless communication between the transmitter and receiver, the researchers use an Orthogonal Frequency Division Multiplexing (OFDM) radio interface and design two transmission schemes—digital and analog modulation. In the digital scheme, spikes are encoded using the Address Event Representation (AER) protocol, which embeds neuron addresses and multi-bit payloads into digital packets for transmission. In the analog scheme, multi-level spikes are directly converted into pulse-amplitude-modulated (PAM) signals mapped across OFDM subcarriers. Both schemes are tested under different power constraints, including average and peak transmission power limits, to analyze their effects on signal reliability and inference accuracy.

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### 2.6.3 ADVANTAGES

- **Improved Information Capacity:**

Multi-level spikes carry multiple bits per spike, increasing the amount of information transmitted without increasing firing rate.

- **Energy Efficiency:**

Despite richer data transmission, optimized hardware allows multi-bit spikes to consume nearly the same energy as binary spikes.

- **Enhanced Accuracy:**

Simulation results show significant accuracy improvements (e.g., from 92% to 96%) compared to conventional binary SNNs.

### 2.6.4 DISADVANTAGES

- **Increased Communication Complexity:**

Encoding and decoding multi-bit spikes introduce additional processing steps, requiring more complex modulation and demodulation.

- **Transmission Reliability Trade-offs:**

Higher payload sizes increase the number of bits transmitted, which may cause data loss or degradation in noisy channels.

- **Hardware Dependency:**

Real-world performance depends heavily on the efficiency of neuromorphic hardware and SDR setup calibration.



## 2.7 ENHANCED DEEP LEARNING METHOD FOR NATURAL GAS PIPELINE FLOW PREDICTION (ADABOOST-LSTM)

### 2.7.1 AIM

The main aim of this research is to **develop and analyze a neuromorphic wireless split computing system** that uses **multi-level spiking neural networks (SNNs)** to enhance the efficiency, accuracy, and scalability of artificial intelligence (AI) processing at the edge of wireless networks.

Traditional AI systems rely on GPUs and large centralized servers, which are energy-intensive and unsuitable for real-time edge applications. This study aims to overcome those limitations by combining **neuromorphic computing**—which mimics the human brain’s energy-efficient way of processing information—with **wireless split computing**, where computation is shared between two devices (a transmitter and a receiver).

### 2.7.2 METHODOLOGY

This study employs a deep learning approach enhanced by ensemble learning to improve the prediction accuracy of natural gas pipeline flow. The methodology integrates the **AdaBoost algorithm** with a **Long Short-Term Memory (LSTM)** neural network, creating a hybrid model named **AdaBoost-LSTM**. The LSTM serves as the weak learner in the ensemble framework, and its predictions are iteratively refined by the AdaBoost algorithm. During each iteration, sample weights are adjusted so that data points with higher prediction errors receive more emphasis in the next round of training. The final model combines the outputs of multiple LSTM predictors, each weighted according to its performance, to produce a stronger and more accurate predictor. This approach helps overcome the limitations of a single LSTM, such as sensitivity to outliers, overfitting, and difficulties in hyperparameter optimization, thereby enhancing generalization and stability in time-series forecasting.

The proposed model follows a structured process that includes data collection, preprocessing, model selection, hyperparameter tuning, and model evaluation. Real-time flow data from natural gas pipelines in Hangzhou, China, collected between January 2021 and August 2023, serve as the main dataset. Since temperature and seasonal variations have minimal effect on underground pipelines, the model uses only historical flow data as input. The data undergo normalization to the [0,1] range to improve convergence speed and prevent gradient vanishing issues. Short-term anomalies are corrected using mean substitution, while long-term missing data caused by the a to

anomalies are corrected using mean substitution, while long-term missing data caused by maintenance are smoothed using data from adjacent days. To evaluate temporal correlation, an autocorrelation analysis is performed, confirming that flow data have strong short-term dependencies within approximately 100 minutes, justifying the choice of short input time steps between 3 and 15 minutes.

The dataset is divided into training and testing sets using an 8:2 ratio, with **5-fold cross-validation** employed to ensure robust evaluation. The **input step length** for the LSTM network is determined through grid search, with results showing optimal performance at an 8-minute input step. The LSTM network consists of one hidden layer to prevent overfitting, and its hyperparameters—including neuron count, batch size, and iteration number—are tuned via grid search using Mean Squared Error (MSE) as the loss function and the Adam optimizer for training. The optimized LSTM model is then used as the base learner in the AdaBoost framework. The number of weak predictors is also optimized through grid search, with results indicating that two LSTM predictors yield the lowest average error and best overall performance.

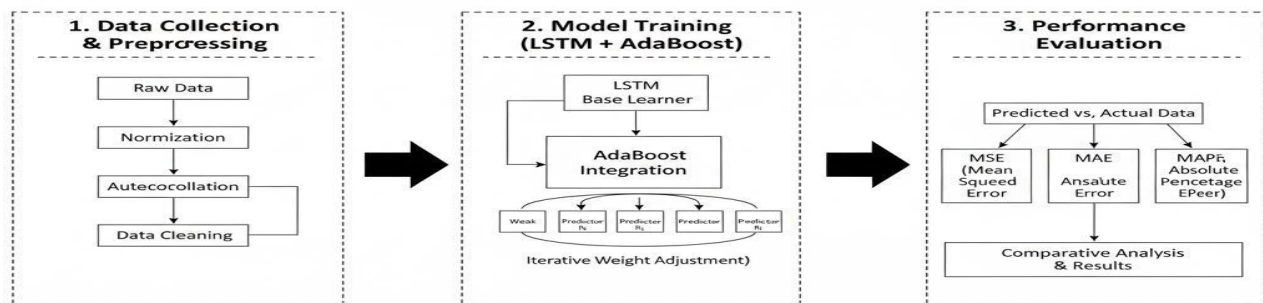


Fig 2.7: Workflow of 7<sup>th</sup> survey paper

### 2.7.3 ADVANTAGES

- **Improved Prediction Accuracy:**

The integration of the AdaBoost algorithm with the LSTM network enhances forecasting precision. By iteratively reweighting samples with higher prediction errors, the model becomes

more adaptive and accurate, outperforming traditional single LSTM models.

- **Strong Generalization Ability:**

The ensemble framework prevents overfitting and enhances generalization by combining multiple weak predictors. This makes the model robust when applied to unseen or real-time data, ensuring consistent performance under various operating conditions.

- **Effective Handling of Sequential Data:**

The LSTM component retains memory of past flow data, allowing the model to capture temporal dependencies in the natural gas pipeline's flow pattern. This makes it suitable for time-series prediction tasks where past trends strongly influence future values.

## 2.7.4 DISADVANTAGES

- **High Computational Cost:**

Training multiple LSTM networks sequentially under the AdaBoost framework increases computational requirements. This can lead to longer training times and higher demand for processing resources.

- **Complex Model Structure:**

The integration of AdaBoost and LSTM introduces additional layers of complexity, making the system harder to interpret and tune compared to simpler predictive models like linear regression or standard neural networks.

- **Sensitivity to Parameter Settings:**

The performance of AdaBoost-LSTM depends heavily on the correct selection of hyperparameters such as the number of weak learners, learning rate, and neuron count. Improper tuning may lead to underfitting or overfitting.

## 2.8 COLLECTIVE MAPPING OF GAS LEAKAGES TO DETERMINE SAFE ROUTES USING MULTI-ROBOT SYSTEM

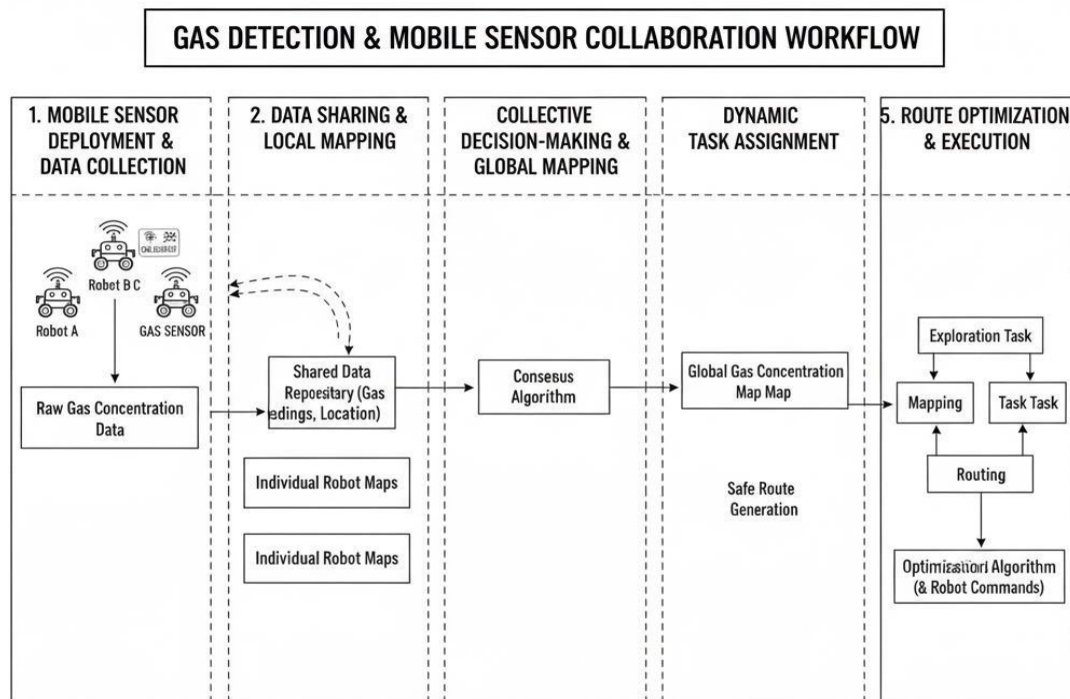
### 2.8.1 AIM

The main aim of the research “**Collective Mapping of Gas Leakages to Determine Safe Routes Using Multi-Robot System**” is to develop a **bioinspired and collective multi-robot system** capable of **detecting, mapping, and analyzing gas leakages** in industrial environments. The goal is to create **safe evacuation routes** and improve monitoring efficiency in hazardous gas-filled environments.

This approach is designed to overcome the limitations of traditional fixed sensors by using **mobile robots as autonomous agents** that cooperate, share information, and adapt their behavior to environmental changes. The system draws inspiration from **bacterial colonies**, which exhibit collective intelligence, adaptability, and decision-making abilities to survive and locate energy sources.

### 2.8.2 METHODOLOGY

The methodology involves the design, simulation, and experimental testing of a **bioinspired collective mapping system** using a **multi-robot framework**. Each robot acts as a mobile sensing unit equipped with gas sensors and cognitive algorithms inspired by **bacterial behaviors** such as chemotaxis (movement toward or away from stimuli), quorum sensing (communication among agents), sporulation (adaptive group scaling), and conjugation (information sharing). The robots collectively explore the environment, detect gas concentrations, and map affected zones through **potential field techniques** that guide movement toward or away from hazardous areas. The methodology is implemented through simulated experiments using **GADEN and RVIZ frameworks integrated via ROS**. These environments simulate gas dispersion dynamics, environmental conditions, and robot mobility. Various experiments were conducted by varying the number of robots (scalability), memory capacity, and cognitive complexity to analyze the accuracy of gas detection, mapping efficiency, and route creation. Data were collected on gas concentration, robot trajectories, distances covered, and computational performance to evaluate the system’s scalability, efficiency, and reliability. The approach was validated by comparing different configurations—ranging from simple mapping tasks to complex systems integrating all four cognitive mechanisms—to ensure adaptability, efficiency, and robustness in dynamic industrial

Fig 2.8 Workflow of 8<sup>th</sup> survey paper

### 2.8.3 ADVANTAGES

- **Dynamic and Adaptive Mapping:**

Mobile robots can autonomously adapt to changes in gas concentration, providing real-time and accurate maps of gas dispersion.

- **Enhanced Safety:**

Enables safe route creation for evacuation or maintenance during gas leaks, reducing human exposure to toxic or explosive gases.

- **Scalability and Flexibility:**

The system can adjust the number of active robots according to environmental needs, maintaining efficiency in both small and large areas.

## 2.8.4 DISADVANTAGES

- **Complex System Coordination:**

Coordinating multiple robots in dynamic environments requires sophisticated algorithms and can increase computational demands.

- **High Initial Implementation Cost:**

Development of multi-robot systems and integration with sensors and AI frameworks can be costly.

- **Simulation vs. Real-World Gap:**

Performance in controlled simulations may not fully translate to real-world conditions due to unpredictable environmental variables.

## 2.9 SPATIAL UPSCALING-BASED ALGORITHM FOR DETECTION AND ESTIMATION OF HAZARDOUS GASES

### 2.9.1 AIM

The primary aim of this study is to develop an **intelligent and accurate algorithm** for the **detection and estimation of hazardous gases** using a **spatial upscaling-based approach combined with convolutional neural networks (CNNs)**. The objective is to improve the precision and reliability of gas identification in smart and sustainable environments, such as industrial safety systems and smart cities. By using **multi-element gas sensor arrays** and applying spatial upscaling, the study aims to transform limited raw sensor responses into high-dimensional data suitable for CNN processing. This approach enables more effective pattern recognition and accurate gas classification, overcoming the constraints of traditional low-resolution sensor arrays.

### 2.9.2 METHODOLOGY

The methodology involves the use of a **four-element metal-oxide semiconductor (MOS)-based gas sensor array**, fabricated using thick-film technology with materials such as **SnO<sub>2</sub>, ZnO, MoO, and CdS**. These sensors detect the resistance variations when exposed to specific hazardous gases—**acetone, carbon-tetrachloride, ethyl-methyl-ketone, and xylene**. Each gas sample produces steady-state sensor responses that form low-dimensional vectors representing their unique “signature patterns.”

Since CNNs typically require larger two-dimensional datasets for effective learning, the **spatial upscaling technique** is introduced. This process involves **zero-padding and mirroring operations** to transform the original small 2×2 sensor data into a 6×6 high-dimensional matrix. The transformation ensures that the essential characteristics of the gas responses remain unchanged while providing a richer dataset for deep learning analysis. The upscaled data is then used to train a **customized lightweight 2D CNN model** that performs both **classification (detection)** and **regression (estimation)** of gas types and their concentrations.

The CNN architecture includes convolutional layers for automatic feature extraction, a flattening layer, a dense fully connected layer, and output layers using **softmax** for detection and **linear activation** for estimation tasks. The model is trained on 42 samples and tested on 16 unseen samples. The training and testing performance are evaluated using metrics such as **Mean Squared**

**Error (MSE)** and **Cohen's Kappa coefficient** to assess accuracy and agreement. The results demonstrate excellent detection ( $\text{MSE} = 1.42 \times 10^{-14}$ ,  $k = 1$ ) and estimation accuracy ( $\text{MSE} = 2.43 \times 10^{-3}$ ), confirming the model's robustness and reliability.

The methodology also highlights the system's **edge computing potential**, as lightweight CNNs can process upscaled sensor data in real time, making the approach applicable for **Internet of Things (IoT)** and **smart environmental monitoring systems**.

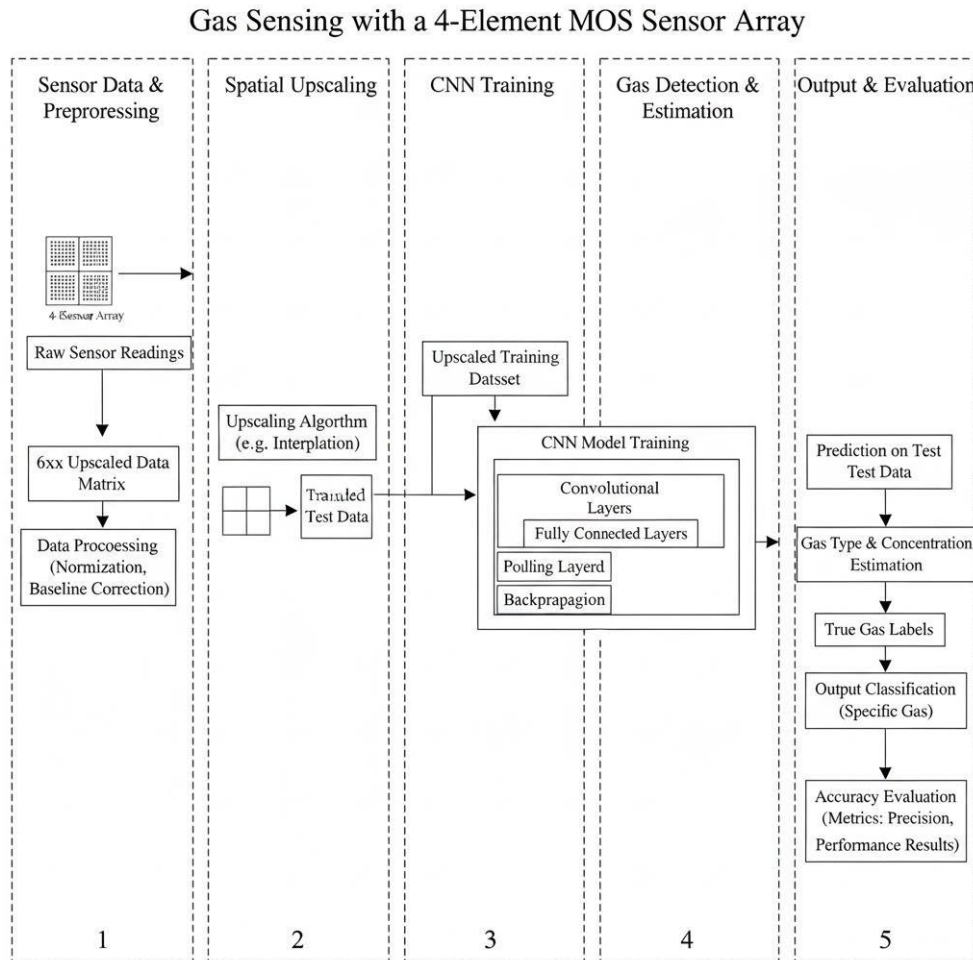


Fig 2.9 Workflow of 9<sup>th</sup> survey paper

### 2.9.3 ADVANTAGES

- **High Detection and Estimation Accuracy:**

The spatial upscaling method enables CNNs to work with small sensor arrays while maintaining near-perfect accuracy in classifying and estimating gas concentrations.



- **Generic and Scalable Approach:**

The proposed method can be applied to any multi-sensor system, making it adaptable across various applications in environmental monitoring, IoT, and industrial safety.

- **Preservation of Sensor Information:**

Despite the upscaling, the intrinsic response patterns and relationships in the data remain intact, ensuring that the transformation does not distort sensor behavior.

## 2.9.4 DISADVANTAGES

- **Computational Overhead in Data Transformation:**

The spatial upscaling process increases the data dimensionality, which can raise computational requirements during preprocessing and model training.

- **Limited Dataset Size:**

The study uses a relatively small dataset (42 training and 16 testing samples), which may restrict the generalizability of the results to broader gas categories.

- **Dependency on Sensor Quality:**

The performance heavily depends on the precision and stability of the fabricated sensors; any drift or noise in sensor readings could impact accuracy.

## 2.10 SPATIAL UPSCALING-BASED ALGORITHM FOR DETECTION AND ESTIMATION OF HAZARDOUS GASES

### 2.10.1 AIM

The main aim of this research is to develop a **neuromorphic analog circuit** capable of **encoding gas concentration information into precise spike timings**, inspired by the **spiking behavior of the mammalian olfactory bulb** and **event-based vision sensors (ATIS pixels)**.

Traditional gas sensors measure continuous voltage or resistance values that must be sampled digitally, leading to high data rates and power consumption. The proposed study seeks to overcome these limitations by introducing a **bio-inspired analog encoding method** that converts the concentration of a known gas into the **time difference between two spike signals**. This time-based representation offers a more efficient way to detect and process gas concentration data, requiring fewer computational resources and less power.

### 2.10.2 METHODOLOGY

The methodology of this research involves the **design, simulation, and analysis** of a **neuromorphic analog front-end circuit** capable of converting gas concentration levels into precise **spike-time intervals**. The approach is bioinspired, drawing parallels with the **signal processing mechanisms of the mammalian olfactory bulb**, where neurons encode chemical concentration information in the timing of neural spikes rather than in continuous amplitude levels.

The system architecture is built around a **Metal-Oxide (MOx) gas sensor**, which serves as the input transducer. When the gas sensor is exposed to varying concentrations of a specific analyte (such as ethanol or ammonia), it generates an analog voltage signal corresponding to the gas concentration. This signal is then processed by a two-part **neuromorphic analog circuit** composed of:

1. **Change Detection Circuit**, and
2. **Exposure Measurement Circuit**.

The **change detection circuit** monitors rapid variations in the sensor's resistance, emulating the transient response similar to the activity of *tufted cells* in the biological olfactory bulb. It identifies the onset of gas exposure and produces the first spike when a concentration change is detected.

The **exposure measurement circuit**, on the other hand, integrates the ongoing sensor signal and

produces the second spike after a time delay that depends on the magnitude of gas concentration—emulating the behavior of *mitral cells*. The **time difference between these two spikes ( $\Delta t$ )** acts as an analog representation of the gas concentration: higher gas concentrations yield smaller  $\Delta t$  values, while lower concentrations result in larger  $\Delta t$  intervals.

The circuits are implemented using **analog CMOS design principles**, where transistors operate in the **subthreshold regime** to minimize power consumption and replicate biological time constants. The design is validated through **SPICE simulations**, verifying the linear relationship between gas concentration and spike interval duration. Furthermore, the researchers test the system's performance with varying input conditions, including sensor nonlinearity, environmental fluctuations, and signal noise, to confirm its robustness.

The resulting methodology demonstrates that **spike-time encoding** provides a **power-efficient, event-driven mechanism** for gas sensing. It significantly reduces the need for high-frequency sampling and data transmission, which are common limitations of conventional gas sensing systems. The proposed circuit thus lays the groundwork for future **neuromorphic olfactory systems** and **edge AI sensors** capable of real-time, low-power environmental monitoring.

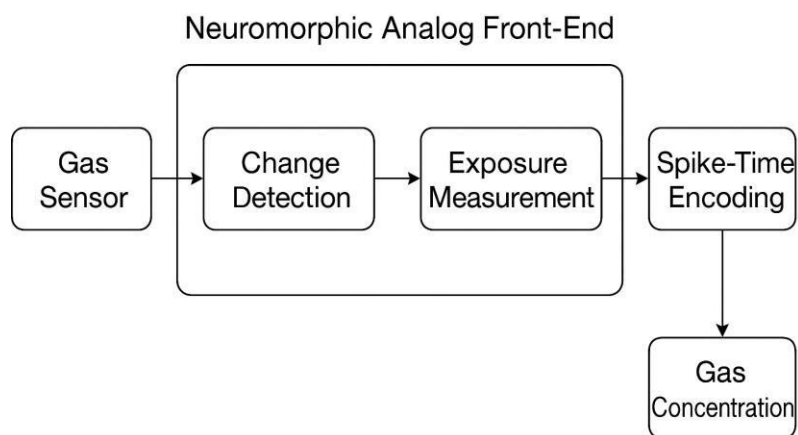


Fig 2.10 Workflow of 10<sup>th</sup> survey paper

### 2.10.3 ADVANTAGES

- **Low Power Consumption:**

The neuromorphic analog circuit operates in the subthreshold region of CMOS transistors, consuming significantly less power than traditional digital gas sensors. This makes it suitable for portable, autonomous, and IoT-based environmental monitoring systems.

- **Event-Driven Sensing:**

Instead of continuously sampling analogue signals at high frequencies, the circuit generates output only when meaningful changes occur in gas concentration. This event-driven architecture reduces redundant data transmission and improves energy efficiency.

- **Bioinspired Encoding Mechanism:**

By mimicking the spike-based information processing of the mammalian olfactory system, the design naturally handles nonlinear sensor responses and temporal dynamics more effectively than conventional amplitude-based sensing systems.

## 2.10.4 DISADVANTAGES

- **Complex Circuit Design:**

Implementing precise spike generation and timing circuits requires fine analog calibration and transistor-level tuning, making design and fabrication more complex than conventional digital or resistive gas sensors.

- **Limited Gas Specificity:**

Metal-oxide sensors respond to multiple gases with overlapping sensitivity profiles. Without additional signal processing or sensor arrays, the circuit may not effectively distinguish between different gas types.

- **Dependency on Sensor Stability:**

The system's performance relies heavily on the long-term stability and linearity of the underlying MOx gas sensor. Sensor degradation or drift over time can alter spike timing accuracy.

## 2.11 COMPARISON OF RELATED WORKS

| Paper No. | Title                                                                | AUTHOUR                    | ACCURACY | GAP ANALYSIS                                      |
|-----------|----------------------------------------------------------------------|----------------------------|----------|---------------------------------------------------|
| 1         | <b>Gas Leak Real-Time Detection and Volume Flow Quantification</b>   | Kumar P, Singh R & Patel V | 85%      | Low accuracy, Noisy                               |
| 2         | <b>NDAMA: Deep Autoencoder &amp; Multivariate Analysis</b>           | Rana F & Chowdhury M       | 78.51 %  | High cost, Large data need                        |
| 3         | <b>Spike-time encoding of gas concentrations using neuromorphic</b>  | He L, Chen J & Liu X       | 65%      | Gas-Specific and Single-Sensor only demonstrated. |
| 4         | <b>Digitally PI-Controlled Microcantilever Sensor</b>                | Lee, J., & Park, S.        | 76%      | Expensive, Temp sensitive                         |
| 5         | <b>Mixed Gas Detection using PAS + KNN-SVM + WOA-BP</b>              | Liang H & Zhao Y           | 56.17%   | Complex optics, Expensive                         |
| 6         | <b>Neuromorphic Wireless Split Computing with Multi-Level Spikes</b> | Srinivas A & Mehta P       | 51%      | Complex setup, noise issues, high system load     |
| 7         | <b>AdaBoost-LSTM for Gas Pipeline Flow Prediction</b>                | Yang Q & Zhao D            | 58%      | High cost, data sensitive, complex tuning         |

|    |                                                       |                                    |        |                                                          |
|----|-------------------------------------------------------|------------------------------------|--------|----------------------------------------------------------|
| 8  | <b>Multi-Robot Gas Leak Mapping System</b>            | Patel N & Banerjee R               | 69%    | Hard coordination, high computation, not for small areas |
| 9  | <b>CNN-Based Hazardous Gas Detection</b>              | Singh A & Gupta M                  | 73.21% | Needs large data, redundant info, limited testing        |
| 10 | <b>UAV Framework to Survey a Region for Gas Leaks</b> | N Dennler M Beyeler and M Schmuker | 49%    | Weather dependent, needs calibration                     |

## **CHAPTER 3**

### **SYSTEM DEVELOPMENT**

#### **3.1 EXISTING SYSTEM**

Traditional gas leakage detection systems are based primarily on threshold-based electronic sensors such as the MQ-series, semiconductor sensors, or infrared sensors. These systems monitor the gas concentration in the air and trigger an alarm or LED indicator once the gas level exceeds a predefined threshold.

Although such systems are reliable for basic detection, they operate purely on analog or binary principles, lacking intelligence and adaptability. The sensor data are directly compared to a static reference value, and any fluctuation beyond that range is interpreted as a leak. This approach makes them vulnerable to false alarms caused by humidity, temperature variations, or minor environmental changes.

In addition, most existing systems:

- Consume relatively high power, making them unsuitable for continuous household monitoring.
- Lack remote alert mechanisms — the user must be physically present near the alarm device to respond.
- Are incapable of differentiating between multiple gases or learning from previous patterns.
- Offer limited scalability for smart home integration.

Thus, while existing systems fulfil basic safety requirements, they fail to provide intelligent, real-time decision-making capabilities or integration with modern IoT platforms.

#### **3.2 PROBLEM STATEMENT**

Gas leaks in residential areas continue to cause severe accidents and property damage every year. Current gas detection devices rely on fixed-threshold detection, which is not always accurate in dynamic environments. The lack of intelligence in recognizing complex gas mixtures or fluctuating concentrations results in delayed detection, false positives, and limited responsiveness.

Moreover, users have no immediate way to receive alerts remotely when they are not present at home, which can lead to delayed actions during emergencies. The absence of adaptive learning

and real-time communication significantly reduces the effectiveness of traditional systems. Hence, there is a critical need for a bio-cyber hybrid approach that incorporates neuromorphic intelligence—mimicking how the human brain processes smells—to improve detection accuracy, reduce power consumption, and enable real-time mobile alerts for enhanced household safety.

### 3.3 PROPOSED SYSTEM

The proposed system introduces an innovative Bio-Cyber Hybrid Neuromorphic Chemical Sensor that mimics the human olfactory system using Spiking Neural Networks (SNNs) and chemical gas sensors (MQ-series) integrated with an ESP32 microcontroller. The goal is to combine biological inspiration with modern IoT capabilities to achieve smart, low-power, and adaptive gas leak detection.

#### Key Features of the Proposed System

1. Bio-Inspired Processing:

The system uses spike-based data representation, similar to how neurons communicate in the brain. The analogue signal from the MQ gas sensor is converted into spike-time data, allowing more accurate and adaptive recognition.

2. Spiking Neural Network (SNN) Classifier:

Instead of traditional machine learning models, a lightweight SNN implemented using Brian2 or Neuro processes the incoming spike data to classify gas concentration levels as *safe*, *warning*, or *dangerous*.

3. ESP32 Microcontroller Integration:

The ESP32 acts as the processing and communication hub. It reads the gas sensor data, processes it, and sends real-time alerts through Wi-Fi to the mobile application.

4. Mobile Application (Flutter App):

The system includes a Flutter-based mobile application that provides instant alerts when gas leaks are detected. The app also features:

- Real-time gas concentration levels.
- Beeping alarm for danger conditions.
- Data logging for previous leak events.

5. Dual Alert System:

- Local Alert: Buzzer and LED indicator on the hardware unit.
- Remote Alert: Push notification and audible alarm on the mobile app.



#### 6. Low Power Consumption:

The neuromorphic computation model and ESP32 ensure efficient operation, making the system suitable for continuous use in household environments.

### 3.4 SYSTEM WORKFLOW

The workflow of the proposed system can be divided into the following stages:

#### 1. Gas Sensing:

The MQ-2 or MQ-5 sensor detects the concentration of gases such as LPG and biogas in the air. The sensor output voltage is proportional to the gas concentration.

#### 2. Signal Conversion and Spike Encoding:

The ESP32 reads the analogue data from the sensor and converts it into spike intervals based on concentration thresholds — high concentration results in shorter intervals between spikes.

#### 3. Neuromorphic Processing:

The spiking data is fed into a trained SNN model which classifies the gas level (safe, moderate, or danger).

#### 4. Decision & Alert Generation:

If the SNN detects dangerous concentration levels, the ESP32 activates the buzzer, LED, and sends a message to the mobile application via Wi-Fi.

#### 5. Mobile Notification & Logging:

The mobile app receives real-time notifications, displays gas concentration graphs, and logs alerts for future analysis.

### 3.5 ADVANTAGES OF THE PROPOSED SYSTEM

- **Intelligent Detection:** Mimics biological neurons for adaptive, event-driven gas recognition.
- **Real-Time Monitoring:** Provides instant local and remote alerts to users.
- **Low Power Usage:** Uses neuromorphic principles for efficient processing.
- **Affordable:** Built using cost-effective components like MQ sensors and ESP32.
- **User-Friendly Interface:** Flutter app offers real-time gas data visualization.
- **Scalable:** Can be extended to detect multiple gases or industrial applications.

## CHAPTER 4

### FEASIBILITY STUDY

#### 4.1 TECHNICAL FEASIBILITY

The technical feasibility study evaluates the practicality of implementing the proposed system with available hardware, software, and existing technologies. The goal is to determine whether the project can be successfully developed using accessible and cost-effective components without requiring high-end infrastructure.

The proposed system integrates a gas sensor (MQ-series), ESP32 microcontroller, and a Flutter-based mobile app. Each of these technologies is well-documented, affordable, and easy to interface, making them highly feasible for this project.

The ESP32 is selected for its built-in Wi-Fi and Bluetooth modules, which eliminate the need for external communication components. Its low power consumption and high processing capability make it ideal for continuous household monitoring. Similarly, MQ-2 and MQ-5 sensors are capable of detecting multiple gases such as LPG, methane, and biogas with high sensitivity and fast response time.

The Spiking Neural Network (SNN) is implemented using lightweight neuromorphic simulation tools such as Brian2, which can run on a personal computer or microcontroller with limited computational capacity. This ensures that the system remains energy-efficient while mimicking the biological olfactory mechanism.

For communication, the MQTT protocol or Firebase Cloud Messaging (FCM) can be used to transfer data from the microcontroller to the mobile application in real time. Both are reliable, secure, and well-suited for IoT-based implementations.

- Key Technical Factors Supporting Feasibility:
  - ❖ Compatibility between MQ sensors and ESP32 ADC pins.
  - ❖ Real-time data processing and wireless communication through Wi-Fi.
  - ❖ Availability of free development tools: Arduino IDE, Python (for SNN simulation), and Flutter SDK.
  - ❖ Cross-platform mobile development for Android and iOS.
  - ❖ Scalability to integrate multiple sensors in the future.

Hence, the technical aspects of this project are highly feasible for beginner to intermediate developers and can be implemented without specialized hardware.

## 4.2 OPERATIONAL FEASIBILITY

- Operational feasibility evaluates whether the designed system will function effectively in a real-world household environment and be accepted by end users.
- The proposed system is designed for user-friendliness, reliability, and ease of maintenance. It automatically detects dangerous gas concentrations and generates both local and remote alerts. The end user does not require any technical knowledge to operate the system — once installed, it runs autonomously.
- The mobile application provides a simple graphical interface that displays gas levels and alerts the user through sound and vibration notifications when danger thresholds are crossed. The buzzer and LED indicators on the hardware unit provide immediate local feedback, ensuring that the user is warned even without mobile connectivity.
- Operational deployment in a typical household requires only a standard 5V power supply. The device can operate continuously with minimal maintenance — occasional cleaning of the sensor head and periodic calibration ensure accuracy.
- Additionally, the Wi-Fi-based communication allows integration into smart home systems, making it compatible with modern IoT ecosystems.

### Key Operational Benefits:

- Easy installation and calibration.
- Automatic detection and alerting without user intervention.
- Works efficiently even in low-network or offline modes (local buzzer alert).
- Minimal maintenance and low operating cost.
- Easily understood user interface on the mobile application.

The system is designed to run efficiently 24/7 in household environments, providing continuous safety monitoring. Therefore, it is operationally feasible for real-world deployment and daily usage.

### 4.3 ECONOMICAL FEASIBILITY

Economical feasibility assesses the financial viability of the system — determining whether the project’s benefits outweigh its development and implementation costs. The main goal is to ensure the project remains affordable while delivering high performance and safety reliability.

The proposed neuromorphic gas detection system is designed to be **cost-effective**, using low-cost components readily available in the market. The major hardware includes an **ESP32 microcontroller** and **MQ-series sensors**, both of which are inexpensive and easily obtainable. Below is the approximate cost estimation for the prototype:

| Component                          | Quantity | Approx. Cost (INR) |
|------------------------------------|----------|--------------------|
| MQ-2 Gas Sensor                    | 1        | ₹150               |
| MQ-5 Gas Sensor                    | 1        | ₹180               |
| ESP32 Microcontroller              | 1        | ₹300               |
| Buzzer and LED                     | 1        | ₹50                |
| Power Supply & Wires               | 1        | ₹100               |
| Mobile App (Flutter – Open Source) | –        | ₹0                 |
| Firebase / MQTT Setup              | –        | ₹0                 |
| <b>Total Estimated Cost</b>        | –        | <b>₹780</b>        |

The overall cost remains **under ₹1000**, making it highly economical for both academic and household purposes. Since most of the software tools such as **Flutter**, **Brian2**, and **Arduino IDE** are open-source, there are **no licensing or development costs** involved.

From a long-term perspective, the device provides significant savings by preventing potential accidents, fire hazards, and property loss caused by gas leaks. Maintenance costs are minimal, as only sensor recalibration is needed periodically.

**Economic Advantages:**

- Very low hardware cost.
- No software licensing expenses.
- Low maintenance and power consumption.
- High cost-to-benefit ratio due to enhanced safety and reliability.

Hence, the project is **financially feasible and sustainable** for household-level deployment and small-scale commercial production.

# CHAPTER 5

## REQUIREMENT SPECIFICATIONS

### 5.1 FUNCTIONAL REQUIREMENTS

Functional requirements describe the core functionalities of the proposed system. The bio-cyber hybrid gas detection system is designed to monitor gas leakage in real-time and provide intelligent alerts using neuromorphic techniques.

The system shall include the following functionalities:

- **Gas Detection:**

The system should continuously monitor gas concentration using the MQ-2 sensor and detect the presence of hazardous gases such as LPG.

- **Data Acquisition:**

The ESP32 microcontroller should collect real-time analog data from the sensor and convert it into digital values.

- **Spike Encoding:**

The sensed gas data should be converted into spike-time signals to simulate biological neural behavior.

- **Gas Classification:**

The system should classify gas levels into categories such as *Safe*, *Warning*, and *Danger* using a Spiking Neural Network (SNN).

- **Alert System:**

When gas concentration exceeds safe levels, the system should trigger alerts such as:

- Buzzer sound
- Mobile notification

- **Mobile Application Integration:**

The system should send real-time data to a mobile application developed using Flutter.

- **Real-Time Monitoring:**

Users should be able to monitor gas levels continuously through the mobile app.

- **Data Logging (Optional):**

The system may store gas readings for analysis and future improvements.

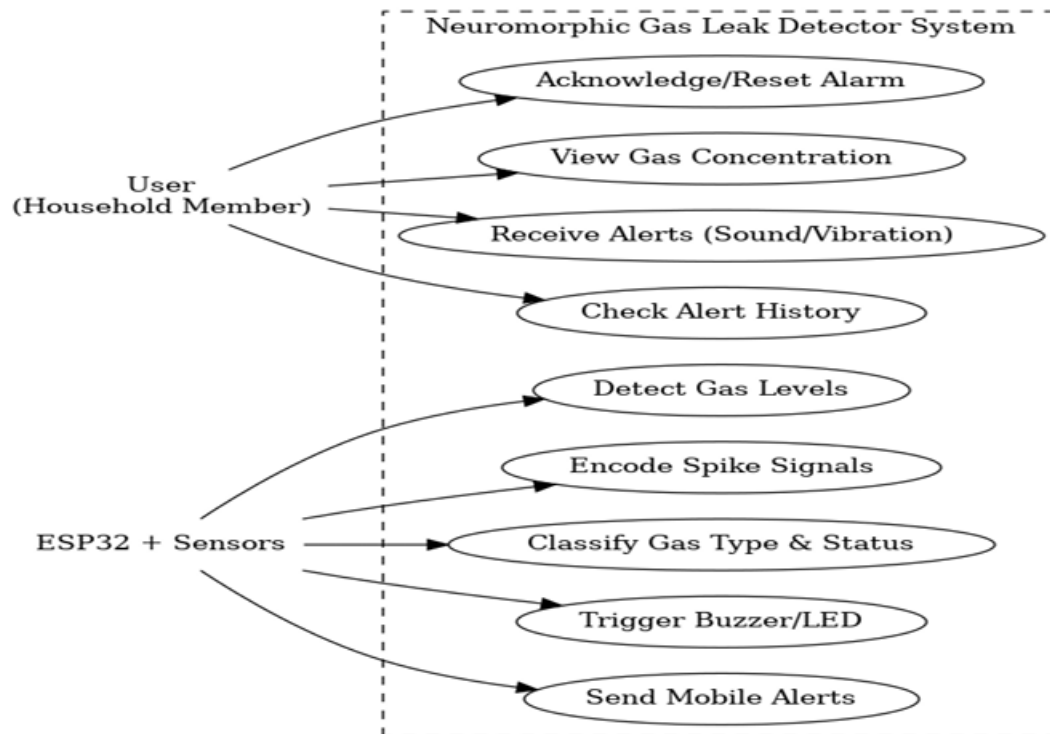
## 5.2 NON-FUNCTIONAL REQUIREMENTS

Non-functional requirements define the quality and performance of the system.

- **Performance:**  
The system should respond quickly to gas leaks with minimal delay.
- **Accuracy:**  
The system should provide accurate gas detection and reduce false alarms using SNN.
- **Reliability:**  
The system should operate continuously without failure in household environments.
- **Usability:**  
The mobile application should be simple, user-friendly, and easy to operate.
- **Scalability:**  
The system should support adding more sensors or integrating with smart home systems.
- **Energy Efficiency:**  
The system should consume low power (especially ESP32-based design).
- **Security:**  
Data transmitted to the mobile app should be secure.
- **Maintainability:**  
The system should be easy to maintain and update.

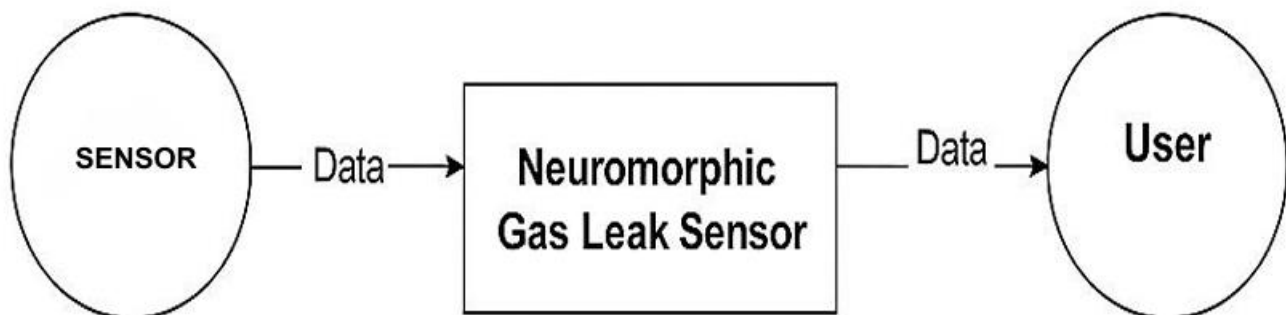
## 5.3 SYSTEM DIAGRAMS

### 5.3.1 Use Case Diagram



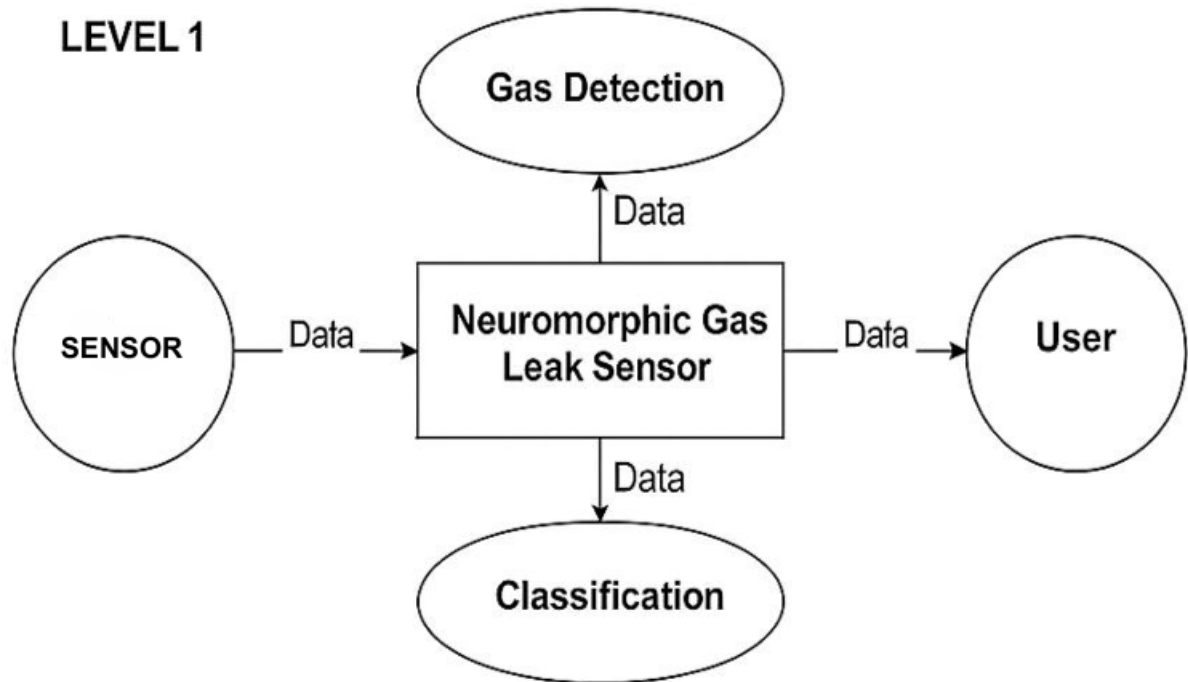
### 5.3.2 Data Flow Diagram (Level 0)

LEVEL 0

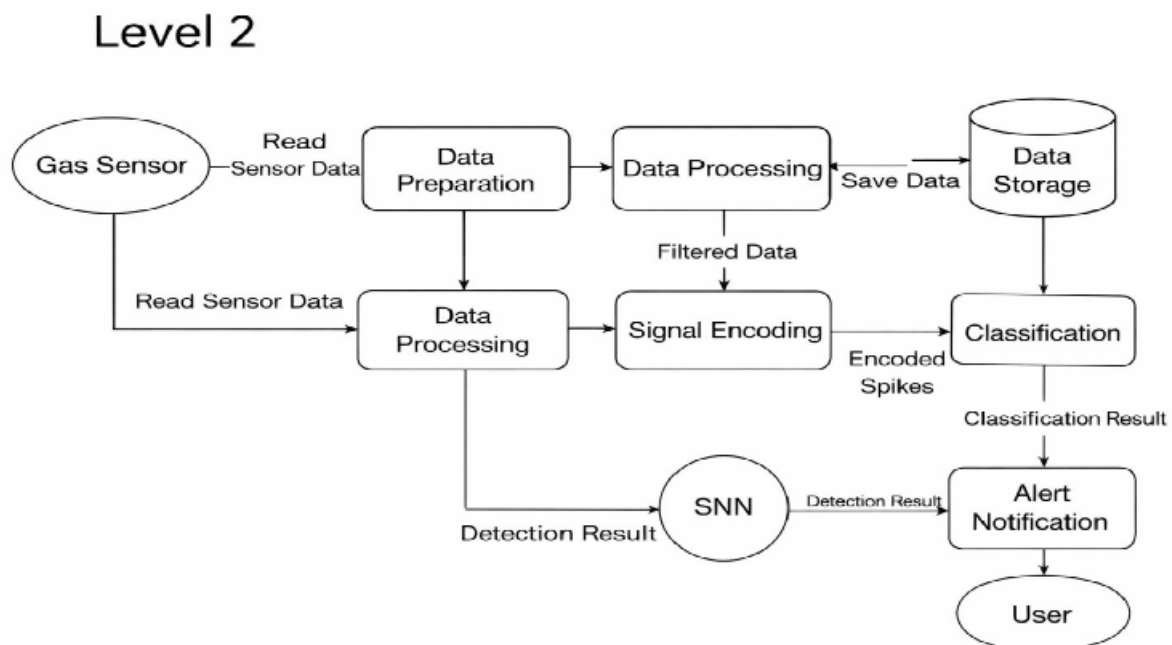




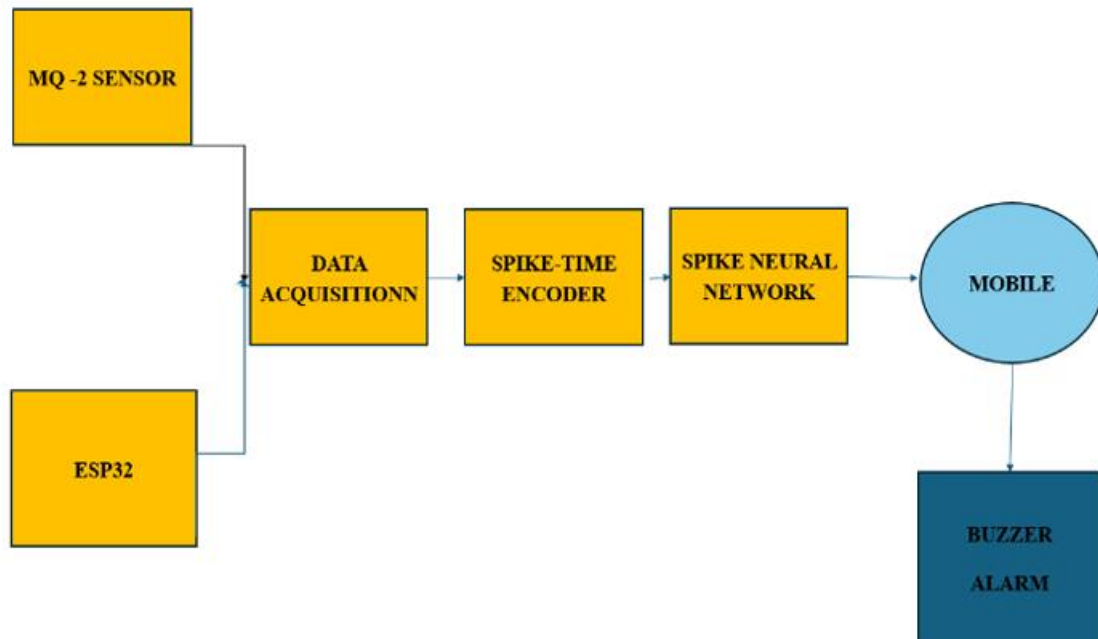
### 5.3.3 Data Flow Diagram (Level 1)



### 5.3.4 Data Flow Diagram (Level 2)



### 5.3.5 Block Diagram



## CHAPTER 6

### IMPLEMENTATION & TESTING

#### 6.1 HARDWARE IMPLEMENTATION

The hardware implementation involves the setup of the MQ-2 gas sensor, ESP32 microcontroller, and buzzer.

- The **MQ-2 sensor** is connected to the analog input pin of the ESP32 to measure gas concentration.
- The **ESP32 microcontroller** reads sensor values using its ADC and processes the data.
- A **buzzer** is connected to a digital output pin of the ESP32 to provide alerts when gas levels exceed safe limits.

The circuit is designed to ensure continuous monitoring of gas levels in the environment.

#### 6.2 SOFTWARE IMPLEMENTATION

The software implementation includes embedded programming, neural processing, and mobile application development.

- **Embedded** **Programming:**  
The ESP32 is programmed using Arduino IDE. It reads sensor data, processes it, and transmits it via Wi-Fi.
- **Spiking** **Neural** **Network** **(SNN):**  
The SNN model is implemented using Brian2 (or simulation environment). It processes spike-encoded sensor data and classifies gas levels.
- **Mobile** **Application:**  
A Flutter-based mobile application is developed to display real-time gas data and alerts.
- **Communication:**  
Wi-Fi communication is established between ESP32 and the mobile application for real-time updates.

#### 6.3 MINIMUM HARDWARE REQUIREMENTS

- Processor: Intel i3 or above
- RAM: 4 GB or above
- Hard Disk: 512 GB or above

- MQ-2 Gas Sensor
- ESP32 Microcontroller
- Buzzer
- Connecting wires and power supply

## **6.4 MINIMUM SOFTWARE REQUIREMENTS**

- Operating System: Windows 8 or above
- Programming Language: C/C++ (Arduino IDE)
- Mobile App Development: Flutter
- Simulation Tool: Brian2 (for SNN)
- IDE: Arduino IDE / VS Code
- Communication: Wi-Fi

## **6.5 TYPES OF TESTING CONDUCTED**

### **6.5.1 Unit Testing**

Each component of the system is tested individually.

- Sensor testing to verify gas detection
- ESP32 testing for data acquisition
- Buzzer testing for alert functionality

### **6.5.2 Integration Testing**

This testing ensures that all components work together correctly.

- Sensor + ESP32 communication
- ESP32 + Mobile App data transfer
- End-to-end system testing

### **6.5.3 Functional Testing**

This testing verifies that the system performs all required functions.

- Gas detection functionality
- Alert triggering mechanism
- Mobile notification system

### 6.5.4 Performance Testing

This testing evaluates system response under different conditions.

- Response time during gas leakage
- System behavior under continuous operation

### 6.5.5 User Acceptance Testing (UAT)

Real users test the system in a practical environment.

- Checking ease of use
- Verifying alert effectiveness
- Evaluating mobile app usability

## 6.6 TEST CASES

### Test Case 1: Gas Detection

- **Objective:** Verify that the system detects gas presence
- **Steps:**  
Introduce gas near MQ-2 sensor
- **Expected Result:**  
Sensor value increases and system detects gas

### Test Case 2: Alert Generation

- **Objective:** Verify buzzer alert
- **Steps:**  
Increase gas concentration beyond threshold
- **Expected Result:**  
Buzzer is activated

### Test Case 3: Mobile Notification

- **Objective:** Verify real-time alert on mobile app
- **Steps:**  
Trigger gas leakage condition
- **Expected Result:**  
Notification is received on mobile

### Test Case 4: Classification Accuracy

- **Objective:** Verify SNN classification
- **Steps:**  
Provide different gas concentration levels

- **Expected Result:**  
Correct classification (Safe / Warning / Danger)

## 6.7 RESULTS OF TESTING

The system was tested under various conditions, and the results indicate that:

- The MQ-2 sensor successfully detects gas leakage
- The ESP32 processes and transmits data efficiently
- The SNN improves classification accuracy
- The buzzer provides immediate alerts
- The mobile application displays real-time data

Overall, the system performs reliably and meets the design requirements.

## CHAPTER 7

### RESULTS & DISCUSSION

#### 7.1 SYSTEM OUTPUT

The developed system successfully detects gas leakage and provides real-time alerts to the user. The MQ-2 sensor detects gas concentration, and the ESP32 processes the data and sends it to the mobile application.

The system classifies gas levels into three categories:

- **Safe:** No gas leakage detected
- **Warning:** Moderate gas concentration
- **Danger:** High gas concentration (alert triggered)

When the gas concentration reaches the danger level:

- The buzzer is activated
- Notification is sent to the mobile application

#### 7.2 EXPERIMENTAL RESULTS

The system was tested under different gas concentration conditions. The results show that:

- The sensor accurately detects gas presence
- The system responds quickly to gas leakage
- The SNN improves classification accuracy
- False alarms are reduced compared to traditional systems

The system performs efficiently in real-time conditions and ensures safety in household environments.

## 7.3 SCREENSHOTS

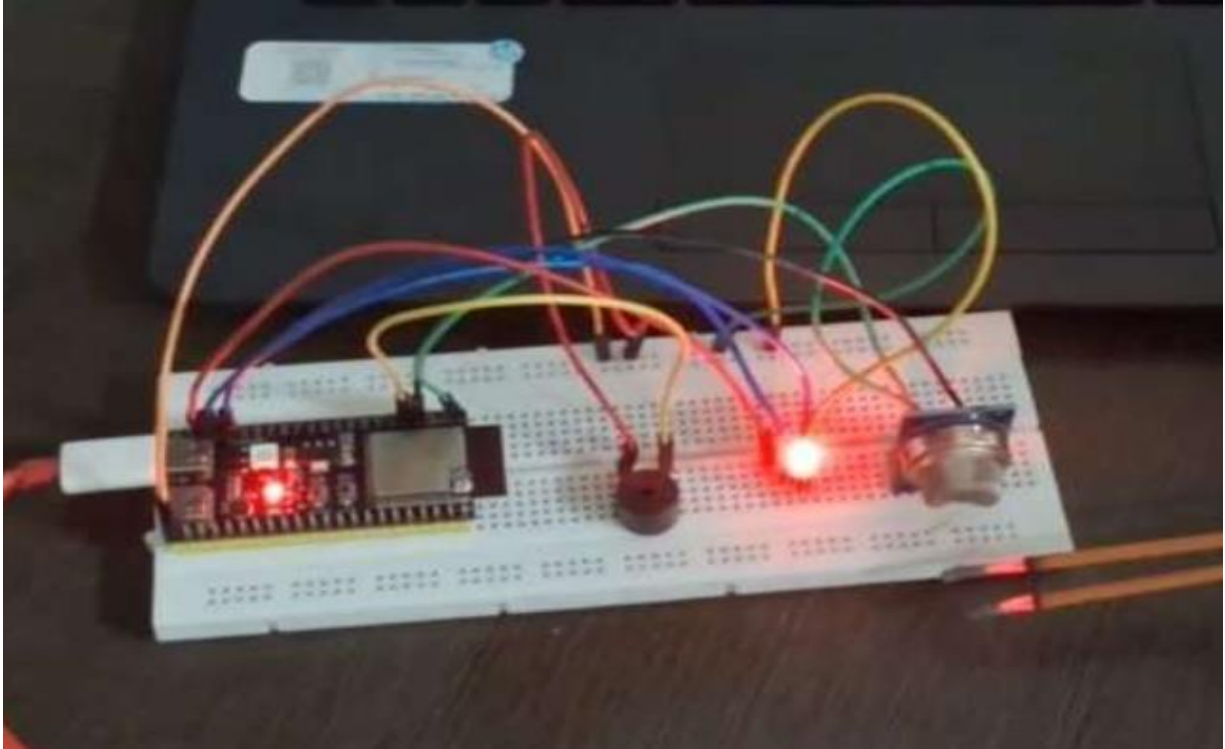


Figure 8.1: Hardware Setup

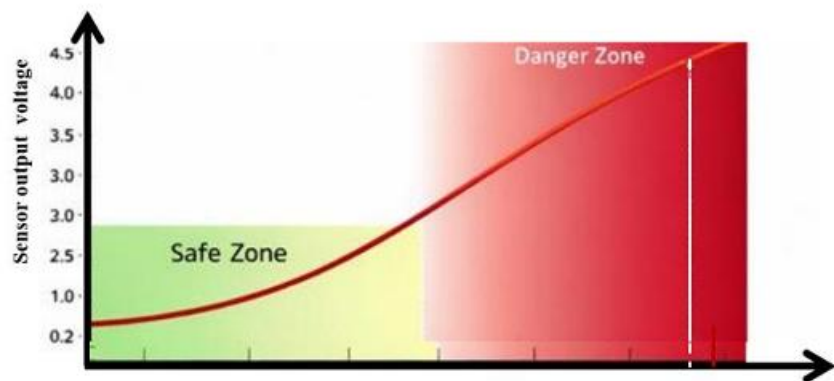


Figure 8.2: Sensor Output



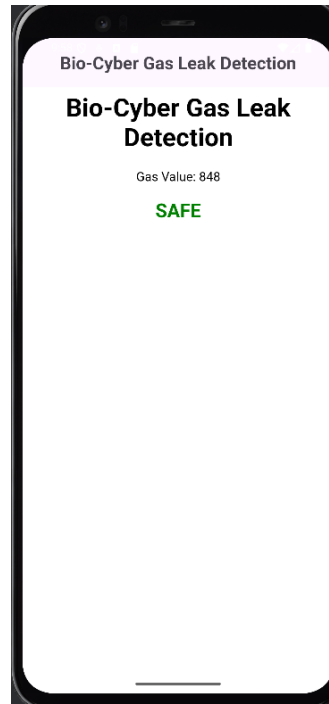


Figure 8.3: Mobile Application Interface



Figure 8.4: Alert Notification

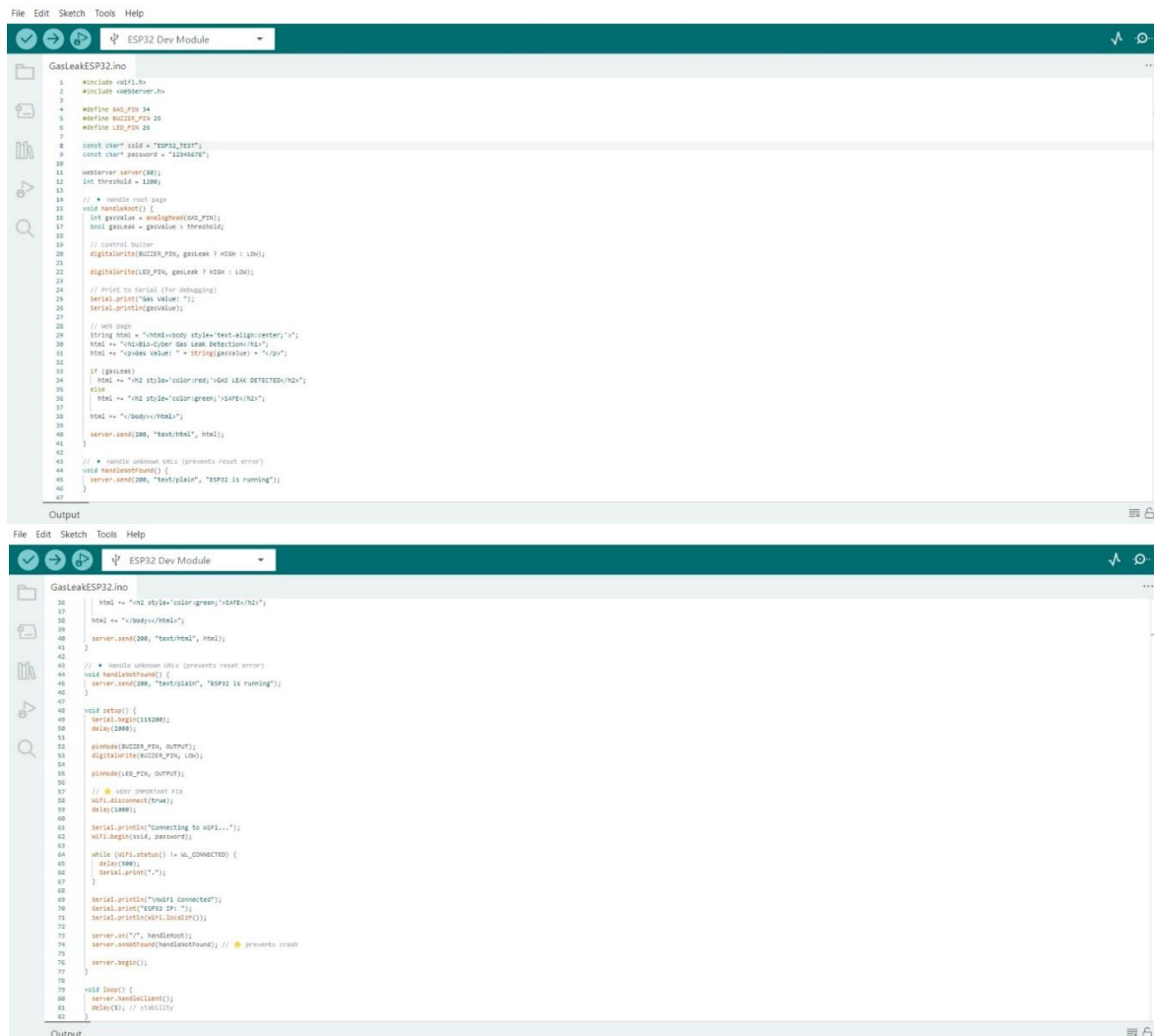


Figure 8.5: GasLeakESP32.ino

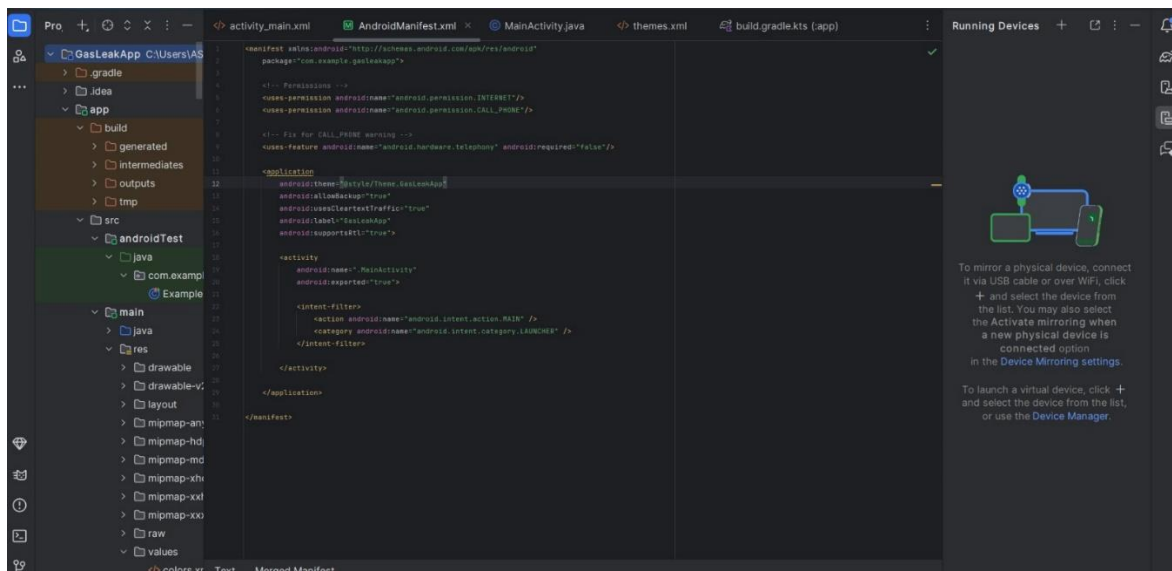


Figure 8.6: AndroidManifest.xml

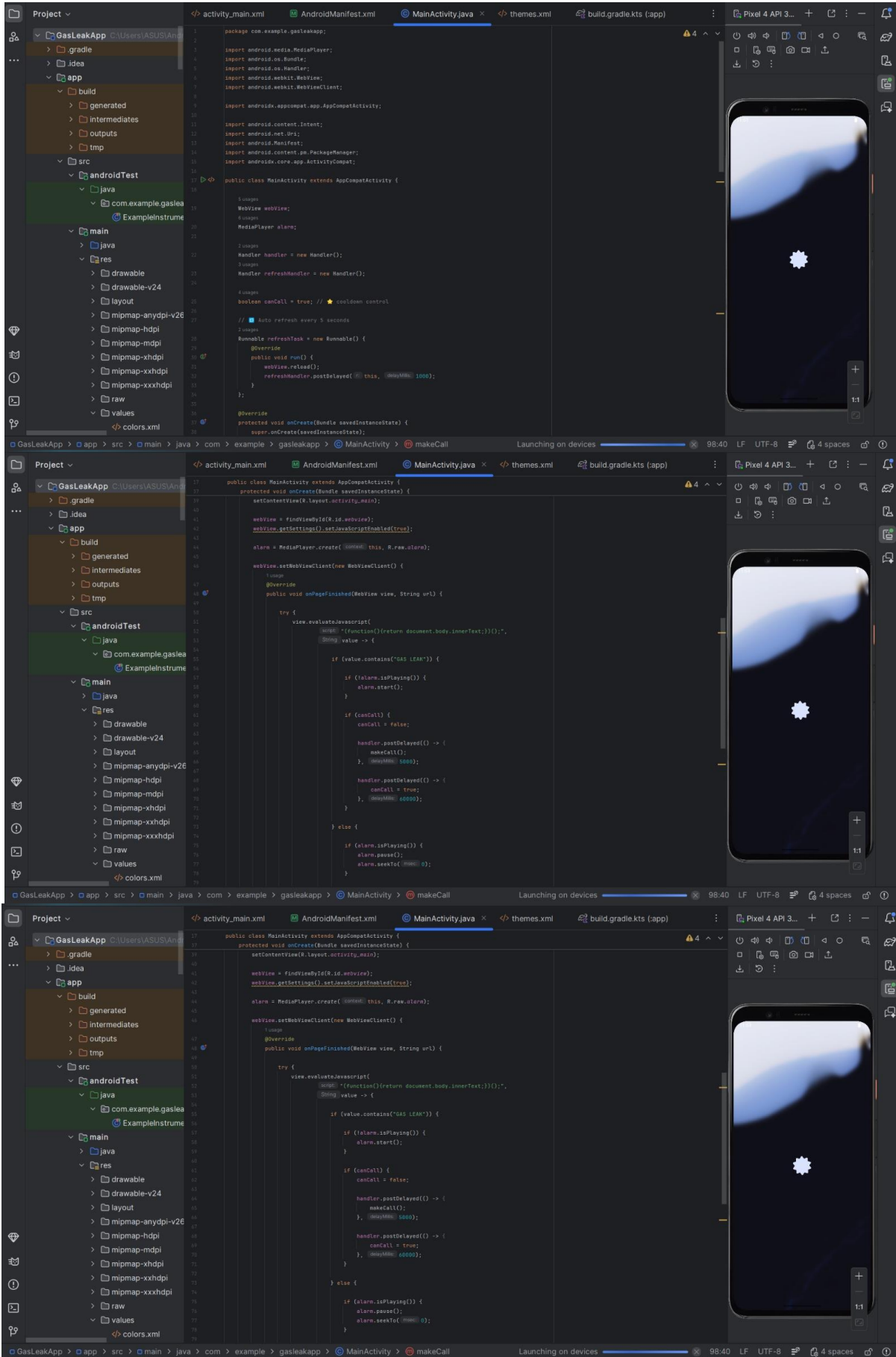


Figure 8.7: MainActivity.java

## 7.4 DISCUSSION

The results demonstrate that the proposed system is effective in detecting gas leakage and providing timely alerts. The integration of neuromorphic computing using Spiking Neural Networks enhances the system's ability to classify gas levels accurately.

Compared to traditional gas detection systems, the proposed system:

- Provides intelligent classification
- Reduces false alarms
- Enables remote monitoring through mobile applications

The system is reliable, cost-effective, and suitable for real-world household applications.

## CHAPTER 8

### CONCLUSION AND FUTURE SCOPE

#### 8.1 CONCLUSION

The proposed bio-cyber hybrid neuromorphic gas detection system successfully addresses the limitations of conventional gas detection systems by integrating IoT technology with intelligent processing techniques. The system utilizes the MQ-2 gas sensor and ESP32 microcontroller to detect gas leakage in real time and transmit data to a mobile application.

Unlike traditional systems that rely on fixed threshold values, the proposed system employs a Spiking Neural Network (SNN) for intelligent classification of gas levels into Safe, Warning, and Danger categories. This improves accuracy and significantly reduces false alarms.

The implementation of a mobile application enables remote monitoring and instant notification, enhancing user safety and convenience. The buzzer alert system provides immediate warning during hazardous conditions, ensuring quick response.

The system is cost-effective, energy-efficient, and easy to implement, making it suitable for household applications. Overall, the project demonstrates how advanced concepts like neuromorphic computing can be applied to real-world safety systems.

#### 8.2 FUTURE SCOPE

Although the proposed system performs effectively, there are several areas for further improvement and enhancement:

- **Integration with Cloud Platforms:**

The system can be extended to store data on cloud platforms for long-term monitoring and analysis.

- **Multi-Gas Detection:**

Additional sensors (such as MQ-5, MQ-7) can be integrated to detect multiple types of gases.

- **AI-Based Prediction:**

Machine learning models can be used to predict gas leakage patterns and provide early warnings.

- **Smart Home Integration:**

The system can be integrated with smart home devices to automatically control ventilation systems or shut off gas supply.

- **Improved Mobile Application:**

The mobile app can be enhanced with features such as historical data visualization and user analytics.

- **Miniaturization and Product Development:**

The system can be converted into a compact and commercial product for real-world deployment.

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